The Portsmith Papers: A Hands-On Tour of PostgreSQL Beyond the Relational Model

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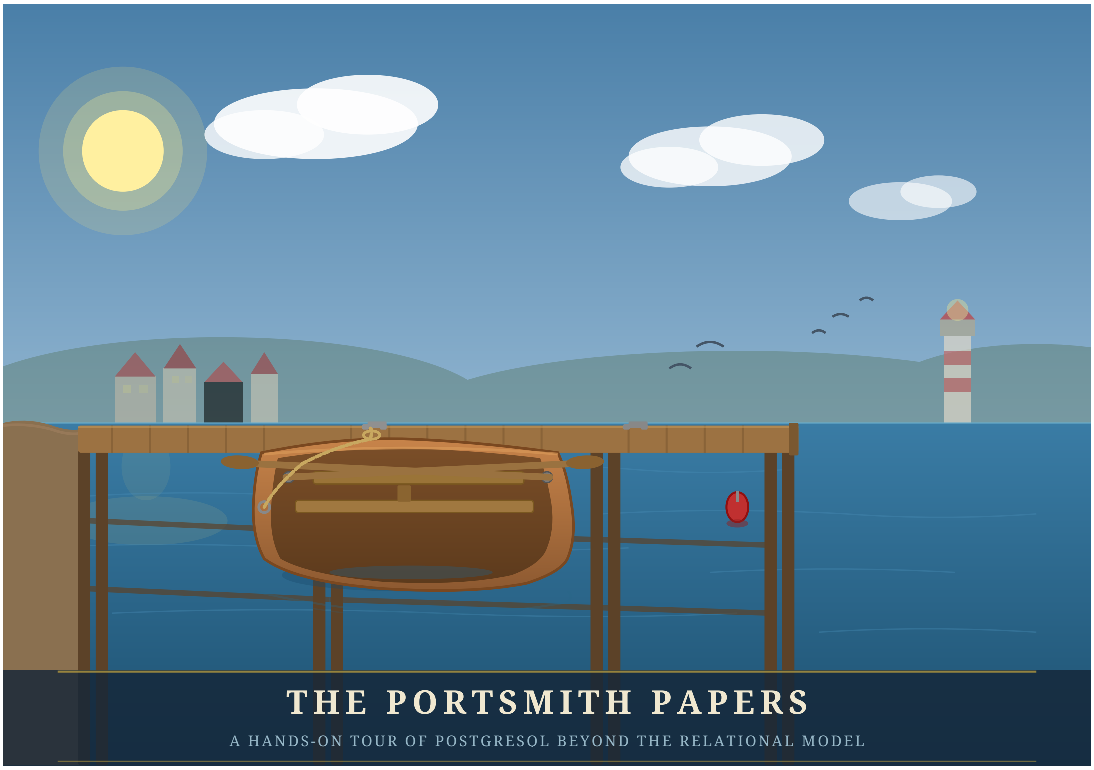

> *"PostgreSQL is not a database with extensions.*
> *It is an extensible data platform that happens to speak SQL."*

About This Book

Most PostgreSQL tutorials end where the interesting work begins.

Once you know how to `SELECT`, `JOIN`, and `GROUP BY`, you have mastered perhaps twenty percent of what PostgreSQL can do. The remaining eighty percent — geospatial queries, semantic search, real-time notifications, fuzzy matching, vector embeddings, distributed coordination, and more — lives in a ecosystem of extensions, index types, and language features that most practitioners never discover.

This book is a guided tour of that other eighty percent.

Each chapter is built around a concrete engineering problem faced by the fictional city of **Portsmith** and its data platform team. We store business directories as semi-structured JSON documents. We route emergency services using geospatial proximity queries. We build a job queue with no message broker. We search municipal records with fuzzy matching that survives typos and OCR errors. We expose the entire platform as a REST API with zero application code.

Every chapter follows the same structure: synthetic data is generated first, giving you a realistic dataset to work against, and then a series of exercises walks you from first principles to production-ready technique. The exercises are written to be done — not just read.

By the end, you will see PostgreSQL not as a place to store rows, but as a programmable data infrastructure layer capable of doing work that most teams reach for separate specialized systems to handle.

What you will need:

- A working PostgreSQL 16 installation (setup covered in Appendix A)
- Python 3.12+
- A Debian-based Linux environment
- Familiarity with basic SQL (`SELECT`, `INSERT`, `JOIN`, `GROUP BY`)
- Curiosity about what else is in there

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Chapter 1 — JSONB: Semi-Structured Data Without a Schema Tax

> *"A schema is a prediction about the future. JSONB lets you hedge."*

Background

Relational databases enforce a contract: every row in a table has exactly the same columns. That contract is usually a feature — it catches mistakes and makes queries predictable. But sometimes you genuinely don't know what columns you'll need until the data arrives. A business directory is a good example. A restaurant has hours, cuisine, and a reservations policy. A hardware shop has trade accounts and parking. A hotel has star ratings and room types. Forcing them all into the same set of columns means drowning in NULLs or building a rats' nest of one-to-many extension tables.

PostgreSQL's `JSONB` type lets you store arbitrary JSON documents in a column while keeping the rest of your row fully relational. It is not a NoSQL escape hatch — it is an indexed, queryable, patchable document field that sits inside a SQL table. You can `WHERE` on it, join against it, aggregate over it, and update individual keys without rewriting the whole document.

This chapter works through all the operators and patterns you'll reach for most often. The exercises are short. Run every one, then try a variation of your own.

The Scenario

Portsmith's city council maintains a public business directory. Every business gets a short relational record — an ID, a name, an address, and a neighbourhood — but the supplementary detail varies so widely by business type that a single fixed schema would be unworkable.

The `businesses` table therefore stores all of that variable metadata in a `details` JSONB column. Each business category contributes its own keys:

Category	JSONB keys present
`restaurant`	`cuisine`, `price_range`, `rating`, `hours`, `tags`, `accepts_reservations`, `outdoor_seating`
`retail`	`subcategory`, `rating`, `hours`, `tags`, `payment_methods`, `has_parking`
`service`	`subcategory`, `rating`, `hours`, `tags`, `appointment_required`, `specialties`
`accommodation`	`subcategory`, `star_rating`, `rating`, `hours`, `amenities`, `room_types`, `price_per_night`
`entertainment`	`subcategory`, `rating`, `hours`, `tags`, `live_music`, `age_restriction`

The `hours` value is itself a nested object keyed by day of week (`mon`–`sun`). Each day is either `null` (closed) or `{"open": "HH:MM", "close": "HH:MM"}`. Several businesses also carry optional keys that appear only for their category, like `social` (social media handles), `happy_hour`, or `approved_brands`.

This heterogeneity is intentional — it is what makes the data a good vehicle for learning JSONB operators.

Exercise Goals

By the end of this chapter you will be able to:

- Extract values from a JSONB document using `->`, `->>`, and `#>>`, and explain why the operator choice changes the result type.
- Filter rows using the containment operator `@>` and the key-existence operators `?`, `?|`, and `?&`.
- Create a GIN index on a JSONB column and confirm that PostgreSQL uses it.
- Update a document in place with `jsonb_set` and the `||` merge operator without rewriting the whole column.
- Expand a JSONB array into a set of rows with `jsonb_array_elements` and aggregate the results.
- Write `jsonb_path_query` expressions to filter on nested values.

Installation

1 — PostgreSQL

If PostgreSQL 16 is not already installed:

```
sudo apt update
sudo apt install -y postgresql-16 postgresql-client-16
```

Confirm the server is running:

```
pg_lsclusters
```

You should see a cluster at version 16 in the `online` state.

2 — Python 3.12 and psycopg

```
sudo apt install -y python3.12 python3.12-venv
```

Create a virtual environment in the book's working directory and install the PostgreSQL driver:

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install "psycopg[binary]"
```

> **Note:** `psycopg` (version 3) is the modern Python driver for PostgreSQL. It is not available as an `apt` package for Python 3.12, so we install it via pip into the virtual environment. The `[binary]` extra bundles a pre-compiled C extension so you don't need `libpq-dev`.

Loading the Data

Create the database

```
# Create a role matching your OS user (skip if it already exists)
sudo -u postgres createuser --createdb "$(whoami)"

# Create the portsmith database
createdb portsmith
```

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch01_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema …
Inserting 48 businesses …
Done — 48 rows in businesses.
```

Verify the load

Open a `psql` session:

```
psql portsmith
```

Run these checks. If all three pass, the data is correct.

Check 1 — Row count and column types:

```
\d businesses
```

```
                                   Table "public.businesses"
    Column     |  Type   | Collation | Nullable |
Default
---------------+---------+-----------+----------
+-----------------------------------
 id            | integer |           | not null |
nextval('businesses_id_seq'::regclass)
 name          | text    |           | not null |
 address       | text    |           | not null |
 neighbourhood | text    |           | not null |
 details       | jsonb   |           | not null |
```

Check 2 — Counts by neighbourhood:

```sql
SELECT neighbourhood, COUNT(*) AS businesses
FROM   businesses
GROUP  BY neighbourhood
ORDER  BY neighbourhood;
```

```
   neighbourhood    | businesses
--------------------+------------
 Harbour District   |          9
 Industrial Port    |          7
 Northgate          |          9
 Old Town           |          9
 Riverside          |          9
```

```
 University Quarter |          5
(6 rows)
```

Check 3 — Counts by category:

```sql
SELECT details->>'category' AS category, COUNT(*) AS businesses
FROM   businesses
GROUP  BY category
ORDER  BY category;
```

```
    category      | businesses
-----------------+------------
 accommodation   |          5
 entertainment   |          6
 restaurant      |         15
 retail          |         12
 service         |         10
(5 rows)
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The Three Extraction Operators

JSONB has two families of accessor. The arrow operators (-> and ->>) navigate one key at a time. The path operator (#>>) jumps straight to a deeply nested value. The difference between them is the **type they return**.

1.1 — Arrow operators

Run these two queries and compare the result types:

```sql
-- -> returns JSONB
SELECT name,
       details -> 'rating' AS rating_jsonb
FROM   businesses
LIMIT  5;

-- ->> returns TEXT
SELECT name,
       details ->> 'rating' AS rating_text
FROM   businesses
LIMIT  5;
```

Look carefully at the column headings in `psql` — one will show `jsonb`, the other `text`. This distinction matters the moment you try to do arithmetic:

```sql
-- This works: cast text to numeric
SELECT name,
       (details ->> 'rating')::numeric AS rating
FROM   businesses
ORDER  BY rating DESC
LIMIT  5;
```

```
          name           | rating
-------------------------+--------
 Lighthouse Bookshop     |    4.9
 Finch & Sons Barbers    |    4.9
 Portsmith Vet. Clinic   |    4.8
 River Bend Bakery       |    4.8
 Quarter Note Jazz Club  |    4.8
(5 rows)
```

> **Key point:** Use `->` when you want to keep working with JSONB (e.g., to navigate deeper). Use `->>` when you need the final value as text for comparisons, casting, or display.

1.2 — The path operator `#>>`

Navigating nested keys with chained `->` quickly becomes unreadable. The path operator takes an array of keys and returns the leaf as text in one step.

```sql
-- Chained arrows (verbose)
SELECT name,
       details -> 'hours' -> 'fri' ->> 'close' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT  8;

-- Path operator (equivalent, cleaner)
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
WHERE  details ->> 'category' = 'restaurant'
LIMIT  8;
```

Both produce the same result. Notice that some rows return `NULL` — those restaurants are closed on Fridays (the `fri` key is JSON `null`).

```
        name         | friday_close
---------------------+--------------
 The Gilded Clam     | 22:30
```

```
 Anchor & Oar Tavern | 01:00
 Bella Napoli        | 23:00
 Le Petit Bistro     | 14:30
 Dragon Palace       | 23:00
 Spice Garden        | 23:00
 Sol y Mar           | 23:00
 Mango Bay Caribbean | 22:30
(8 rows)
```

Exercise 2 — Filtering with Containment and Key Existence

2.1 — Containment: `@>`

The containment operator checks whether the left JSONB document contains all the key-value pairs in the right document. It is the most common way to filter on JSONB values.

Find all waterfront businesses:

```sql
SELECT name, neighbourhood
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}'
ORDER  BY neighbourhood, name;
```

```
          name           |  neighbourhood
-------------------------+------------------
 Anchor & Oar Tavern     | Harbour District
 Harbour Inn             | Harbour District
 Harbour View Theater    | Harbour District
 Mariners Rest B&B       | Harbour District
 Portsmith Fish Market   | Harbour District
 Saltbox Gallery         | Harbour District
 The Gilded Clam         | Harbour District
 Tidal Wave Surf Shop    | Harbour District
 Old Brewery Tap         | Industrial Port
 The Rusty Anchor        | Industrial Port
(10 rows)
```

> **Why this works:** The `@>` operator checks whether the tags *array* on the left contains the tags array `["waterfront"]` on the right — array containment is supported natively.

2.2 — Key existence: `?`

The `?` operator tests whether a key exists in a JSONB object (or a value exists in a JSONB array). Unlike `@>`, it does not check the value — only presence.

Find all businesses that publish social media handles:

```
SELECT name,
       details -> 'social' AS social_links
FROM   businesses
WHERE  details ? 'social'
ORDER  BY name;
```

```
          name          |
social_links
------------------------
+------------------------------------------------------------
 Bella Napoli           | {"instagram": "@bellanapoli_portsmith"}
 Le Petit Bistro        | {"instagram": "@lepetitbistro"}
 Quarter Note Jazz Club | {"instagram": "@quarternote_portsmith"}
 Spice Garden           | {"instagram": "@spicegarden_portsmith"}
 The Gilded Clam        | {"facebook": "thegildedclam",
"instagram": "@gildedclam"}
 The Riverside Vegan    | {"instagram": "@riverside_vegan"}
(6 rows)
```

2.3 — Any-key and all-key existence: `?|` and `?&`

`?|` returns true if *at least one* of the listed keys exists. `?&` returns true only if *all* listed keys exist.

```
-- Businesses that offer either delivery OR takeaway
SELECT name
FROM   businesses
WHERE  details ?| ARRAY['delivery', 'takeaway']
ORDER  BY name;

-- Businesses that offer BOTH delivery AND takeaway
SELECT name
FROM   businesses
WHERE  details ?& ARRAY['delivery', 'takeaway']
ORDER  BY name;
```

Run both and note the difference in result count. The `?&` set is a strict subset of the `?|` set.

Exercise 3 — Missing Keys vs. NULL Values

JSONB has two distinct ways to represent “no value”: a JSON `null` and a **missing key**. They behave differently in queries.

In our data, a business closed on Monday can be represented two ways:

- `"mon": null` — the key exists, the value is JSON null
- the `mon` key is absent entirely

The seed data uses `"mon": null` for closures, so we can observe this.

3.1 — The difference matters for `?`

```
-- ? checks key EXISTENCE, not value
SELECT name
FROM   businesses
WHERE  (details -> 'hours') ? 'sun'    -- key exists (even if null)
ORDER  BY name;
```

Run it. You should get **43 rows** — every business that uses a day-keyed hours object. The five accommodation businesses are excluded because their hours look like `{"reception": "07:00-23:00"}` and have no `sun` key.

Now try what seems like a natural follow-up:

```
--   This does NOT do what you might expect
SELECT name
FROM   businesses
WHERE  (details -> 'hours' -> 'sun') IS NOT NULL
ORDER  BY name;
```

Run it. You still get **43 rows** — identical to the first query.

Why? This is one of JSONB's most important gotchas:

> **JSONB `null` is not SQL `NULL`.**

When the seed script stores `"sun": None` in Python, `json.dumps` produces `"sun": null` — a JSON null *value* under an existing key. When PostgreSQL evaluates `details -> 'hours' -> 'sun'` on such a row, it returns the JSONB value `null`. That is a real value — not the absence of a value — so `IS NOT NULL` evaluates to `TRUE`.

SQL `NULL` only appears when the key itself is **missing** from the object. That is what happens for the five accommodation businesses: `details -> 'hours'` returns `{"reception": "..."}`, and then `-> 'sun'` on that object finds no such key and returns SQL `NULL`, so `IS NOT NULL` is `FALSE`.

To correctly distinguish the three states, use `jsonb_typeof()`:

State	`details -> 'hours' -> 'sun'`	`jsonb_typeof(...)`	`IS NOT NULL`
Key missing (accommodation)	SQL `NULL`	SQL `NULL`	`FALSE`
Key present, closed (`null`)	JSONB `null`	`'null'`	**`TRUE`** ← gotcha
Key present, open (object)	JSONB object	`'object'`	`TRUE`

The correct query for "businesses actually open on Sunday":

```
SELECT name
FROM   businesses
WHERE  jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
ORDER  BY name;
```

This returns **25 rows** — only the businesses with a real hours object for Sunday. `jsonb_typeof` returns `'object'` only for JSON objects, `'null'` for JSON null, and SQL `NULL` for a missing key (which then fails the = test).

> **Rule of thumb:** Never use `IS NOT NULL` to test whether a JSONB navigation result is a "real" value. Use `jsonb_typeof(...)` to check the actual JSON type, or navigate one level deeper (e.g., check `details #>> '{hours,sun,open}'`) — the #>> operator converts JSON null to SQL NULL, so a deeper path naturally filters it out.

3.2 — Handling accommodation differently

Hotels and hostels store hours as a single `"reception": "HH:MM-HH:MM"` string rather than a day-keyed object. This means the `? 'sun'` test returns false for them even though they may be open 24/7. Write a query that finds all businesses that are either:

- Open on Sunday (day-keyed hours with a non-null `sun`), **or**
- An accommodation with 24/7 reception

```
SELECT name, neighbourhood,
       CASE
           WHEN jsonb_typeof(details -> 'hours' -> 'sun') =
'object'
               THEN (details #>> '{hours,sun,open}') || '–' ||
                    (details #>> '{hours,sun,close}')
           WHEN details #>> '{hours,reception}' = '24/7'
               THEN '24/7'
           ELSE details #>> '{hours,reception}'
```

```sql
       END AS sunday_availability
FROM   businesses
WHERE  jsonb_typeof(details -> 'hours' -> 'sun') = 'object'
   OR  (details -> 'hours') ? 'reception'
ORDER  BY neighbourhood, name;
```

Exercise 4 — GIN Indexes

Without an index, every JSONB query is a sequential scan — PostgreSQL reads every row and evaluates the expression. For a 48-row toy dataset that is unnoticeable, but with millions of rows it becomes a bottleneck.

A **GIN** (Generalised Inverted Index) index over a JSONB column indexes every key and value inside every document. It accelerates `@>`, `?`, `?|`, and `?&` queries.

4.1 — Observe the plan before indexing

```sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';
```

With 48 rows you will see `Seq Scan` (sequential scan, i.e. reading every row) — that is expected. On a larger table this would say `Seq Scan` with a high cost estimate.

4.2 — Create the GIN index

```sql
CREATE INDEX idx_businesses_details_gin
    ON businesses
    USING GIN (details);
```

4.3 — Observe the plan after indexing

```sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';
```

On a small table PostgreSQL may still choose a sequential scan (the planner knows the table fits in memory). Force it to use the index to see the mechanism:

```sql
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
```

```sql
FROM   businesses
WHERE  details @> '{"tags": ["waterfront"]}';

SET enable_seqscan = on;   -- always restore this
```

You should now see `Bitmap Index Scan on idx_businesses_details_gin` in the plan. The GIN index will pay for itself in production as the table grows.

> **What GIN indexes:** The default `jsonb_ops` operator class indexes every key and value in the document, supporting `@>`, `?`, `?|`, `?&`. A `jsonb_path_ops` class indexes only values (not keys), producing a smaller index that is faster for `@>` but cannot accelerate `?` queries. Use `jsonb_path_ops` when you only ever query by value containment and index size matters.

```sql
-- Alternative: value-only index (smaller, faster @>, no ? support)
CREATE INDEX idx_businesses_details_pathops
    ON businesses
    USING GIN (details jsonb_path_ops);
```

Exercise 5 — Updating Documents in Place

JSONB columns are immutable at the document level — you cannot change a single key without rewriting the whole value. PostgreSQL gives you two tools to do that rewrite concisely.

5.1 — `jsonb_set`: replace or add one value

The Lighthouse Bookshop just received a flood of five-star reviews. Update its rating to 5.0:

```sql
UPDATE businesses
SET    details = jsonb_set(details, '{rating}', '5.0')
WHERE  name = 'Lighthouse Bookshop';
```

Verify:

```sql
SELECT name, details ->> 'rating' AS rating
FROM   businesses
WHERE  name = 'Lighthouse Bookshop';
```

`jsonb_set(target, path, new_value)` — the path is a text array of keys. If the key already exists it is replaced; if it does not exist it is added (by default — there is a fourth parameter to suppress creation).

5.2 — `jsonb_set` on a nested key

The Gilded Clam is now closing an hour later on Sundays:

```sql
UPDATE businesses
SET    details = jsonb_set(
                    details,
                    '{hours,sun,close}',
                    '"21:00"'     -- note: a JSON string must be double-quoted
                )
WHERE  name = 'The Gilded Clam';

SELECT name,
       details #>> '{hours,sun,close}' AS new_sunday_close
FROM   businesses
WHERE  name = 'The Gilded Clam';
```

5.3 — `||` merge operator: add or overwrite multiple keys at once

The `||` operator merges two JSONB objects, with the right side winning on conflicts. Use it to add a `verified` flag and update `review_count` in a single statement:

```sql
UPDATE businesses
SET    details = details || '{"verified": true, "review_count": 999}'
WHERE  name = 'Anchor & Oar Tavern';

SELECT name,
       details ->> 'verified'     AS verified,
       details ->> 'review_count' AS reviews
FROM   businesses
WHERE  name = 'Anchor & Oar Tavern';
```

> **Caution:** `||` does a *shallow* merge. If `details` contains a nested object like `hours` and you merge another object that also has `hours`, the entire `hours` value is replaced — not deep-merged. Use `jsonb_set` for targeted nested updates.

Exercise 6 — Expanding Arrays with `jsonb_array_elements`

The `tags` key in most business documents is a JSON array. To query across tags — for example, to find the most common ones across the whole directory — you need to expand the array into individual rows.

6.1 — Expand one business's tags

```sql
SELECT name,
       jsonb_array_elements_text(details -> 'tags') AS tag
```

```sql
FROM   businesses
WHERE  name = 'The Gilded Clam';
```

```
       name       |      tag
------------------+---------------
 The Gilded Clam  | waterfront
 The Gilded Clam  | seafood
 The Gilded Clam  | romantic
 The Gilded Clam  | outdoor_seating
(4 rows)
```

> `jsonb_array_elements_text` is a set-returning function: each element of the array becomes its own row. The result has more rows than the input.

6.2 — Count tags across all businesses

```sql
SELECT tag, COUNT(*) AS occurrences
FROM   businesses,
       jsonb_array_elements_text(details -> 'tags') AS tag
GROUP  BY tag
ORDER  BY occurrences DESC
LIMIT  15;
```

```
       tag        | occurrences
------------------+------------
 waterfront       |         10
 takeaway         |          5
 family_friendly  |          4
 family_owned     |          4
 organic          |          3
 ...
(15 rows)
```

Your exact numbers will differ — this shows the query pattern. The comma between `businesses` and the `jsonb_array_elements_text(...)` call is a **lateral join** (PostgreSQL expands the function for each input row automatically when used in the `FROM` clause this way).

6.3 — Only businesses with multiple specific tags

Find restaurants that are both `vegan_options` and `takeaway`:

```sql
SELECT name, details ->> 'cuisine' AS cuisine
FROM   businesses
WHERE  details @> '{"tags": ["vegan_options"]}'
  AND  details @> '{"tags": ["takeaway"]}';
```

Because each @> call is independent, both conditions must hold. This is more efficient than expanding the array when the GIN index is in place.

Exercise 7 — Path Queries with `jsonb_path_query`

The `@?` operator and `jsonb_path_query()` function implement **SQL/JSON path language** — a mini-query language for navigating and filtering within a JSONB document. It is more expressive than chained arrow operators for conditional navigation.

7.1 — Simple path existence

Find businesses where the social links include Instagram:

```
SELECT name
FROM   businesses
WHERE  details @? '$.social.instagram';
```

`$.social.instagram` means: start at the root (`$`), descend to `social`, then to `instagram`. `@?` returns true if the path resolves to at least one value.

7.2 — Filter on a nested value

Find restaurants with a rating above 4.5:

```
SELECT name,
       details ->> 'rating' AS rating
FROM   businesses
WHERE  details @? '$ ? (@.category == "restaurant" && @.rating >
4.5)';
```

```
         name          | rating
-----------------------+--------
 Bella Napoli          | 4.6
 Le Petit Bistro       | 4.7
 River Bend Bakery     | 4.8
 Spice Garden          | 4.6
 The Gilded Clam       | 4.5
(5 rows)
```

The `? (filter)` syntax inside a path expression is equivalent to a `WHERE` clause inside the document.

7.3 — Find businesses open late on Friday

Businesses that close at or after 21:00 on Friday (using string comparison, which works correctly for 24-hour `HH:MM` strings):

```
SELECT name,
       details #>> '{hours,fri,close}' AS friday_close
FROM   businesses
```

```
        WHERE  details @? '$.hours.fri.close ? (@ >= "21:00")'
        ORDER  BY friday_close DESC, name;
```

```
          name            | friday_close
--------------------------+--------------
 Bella Napoli             | 23:00
 Dragon Palace            | 23:00
 Harbour View Theater     | 23:00
 The Hungry Scholar       | 23:00
 Spice Garden             | 23:00
 Sol y Mar                | 23:00
 Mango Bay Caribbean      | 22:30
 The Gilded Clam          | 22:30
 Thai Orchid              | 22:00
 The Riverside Vegan      | 22:00
 Northgate Grocers        | 21:00
 Lotus Spa & Wellness     | 20:00
(12 rows)
```

> **Limitation to notice:** Businesses that close *after midnight* — Anchor & Oar, The Clocktower Pub, The Rusty Anchor, and others — show a `close` of `"01:00"` or `"02:00"`. Lexicographically, `"02:00" < "21:00"`, so these businesses are *excluded* from the above results even though they are open very late. This is a design trade-off of storing times as plain strings. One solution is to store closing times past midnight as `"25:00"`, `"26:00"`, etc. — an unusual but practical convention for 24-hour time arithmetic.

7.4 — Extract values with `jsonb_path_query`

`@?` is a boolean test. `jsonb_path_query()` returns the actual matched values:

```
        SELECT name,
               jsonb_path_query(details, '$.hours.fri.close') AS
friday_close_json
        FROM   businesses
        WHERE  details @? '$.hours.fri'
        ORDER  BY name
        LIMIT  8;
```

The returned values are JSONB (note the quotes around the time strings). Use `jsonb_path_query_first(...) #>> '{}'` to extract a single scalar as text without the surrounding quotes.

Summary — What You Should Now Know

You have worked through the core JSONB toolkit. Here is what each operator and function you used actually does:

Tool	What it does
`-> 'key'`	Navigate one level; returns **JSONB**
`->> 'key'`	Navigate one level; returns **text**
`#>> '{a,b,c}'`	Navigate a path; returns **text**; JSON null becomes SQL NULL
`@> '{"k":"v"}'`	Containment test; accelerated by GIN
`? 'key'`	Key exists test (regardless of value); accelerated by GIN
`?|` / `?&`	Any-key / all-keys exist; accelerated by GIN
`jsonb_typeof(val)`	Returns `'object'`, `'array'`, `'string'`, `'number'`, `'boolean'`, or `'null'`; SQL NULL for a missing key
`jsonb_set(col, path, val)`	Replace or insert a nested value
`col || '{"k":"v"}'`	Shallow-merge a document
`jsonb_array_elements_text(col)`	Expand a JSON array into rows
`@? 'path'`	SQL/JSON path existence test
`jsonb_path_query(col, 'path')`	SQL/JSON path — return matched values

> **Remember:** JSONB `null` ≠ SQL `NULL`. Use `jsonb_typeof()` — not `IS NOT NULL` — when you need to distinguish a missing key from a key that exists but holds a JSON null value.

The key design insight from this chapter is that JSONB lets you keep a relational spine — the columns your queries always need (`id`, `name`, `neighbourhood`) — while storing everything else in a document whose shape can vary row by row. The GIN index means you do not pay a query performance penalty for that flexibility.

In the next chapter you will add point geometry to the `businesses` table and use PostGIS to answer spatial questions: which businesses are within 500 metres of the harbour, and which neighbourhood does each one belong to.

Going further: PostgreSQL 14+ supports the `jsonpath` type natively. The `jsonb_path_query_array` and `jsonb_path_query_first` variants are useful for pagination and single-value extraction. For write-heavy workloads, *profile whether `JSONB` or a computed stored column (Chapter 16) gives better `INSERT`/`UPDATE` throughput on your hardware.*

Chapter 2 — PostGIS: Geospatial Queries on Real Geometry

> *“A city is not a list of rows. It is a shape on the ground.”*

Background

Every interesting question about a city is ultimately a spatial question. Which businesses are near the harbour? Which neighbourhood is this address in? How large is the industrial waterfront? Relational databases answer these badly when location is stored as a text field or a pair of float columns — there is no native concept of “within”, “contains”, or “distance”.

PostGIS is a PostgreSQL extension that adds first-class geometry and geography types, plus several hundred spatial functions and operators. It turns PostgreSQL into a full spatial database: you can store points, lines, and polygons; index them with GIST; and answer proximity, containment, and area queries in SQL without an external GIS system.

This matters in practice far beyond mapping applications. Address geocoding, logistics routing, fraud detection (is this login coming from the expected region?), real estate valuation, and urban planning all reach for spatial queries. PostGIS is the standard tool for all of them in the PostgreSQL ecosystem.

The Scenario

The Portsmith business directory from Chapter 1 stores each business’s neighbourhood as a plain text column. That works for simple filtering, but it cannot answer *where* questions: it cannot find businesses near a given coordinate, cannot verify that a business address actually falls inside its declared neighbourhood, and cannot measure distances.

This chapter adds a point geometry to every business record, then introduces three new spatial tables:

Table	Geometry type	What it holds
`neighborhoods`	`POLYGON`	Boundary polygons for Portsmith's six neighbourhoods
`parks`	`POLYGON`	Six public parks and green spaces
`city_infrastructure`	`LINESTRING`	Twelve named road segments

All coordinates are in WGS-84 (SRID 4326), the same coordinate system used by GPS and most web mapping APIs.

Exercise Goals

By the end of this chapter you will be able to:

- Store and inspect `POLYGON` geometry using WKT (Well-Known Text) and `ST_GeomFromText`.
- Run proximity searches with `ST_DWithin`, and understand why the `::geography` cast matters for distance accuracy.
- Perform spatial joins using `ST_Within` and `ST_Contains`, and explain the difference between them.
- Compute polygon areas in square kilometres using `ST_Area` with the geography type.
- Find the nearest feature for every row using `ST_Distance` and a `CROSS JOIN LATERAL`.
- Create a GIST spatial index and confirm that PostgreSQL uses it.

Installation

1 — PostGIS server package

PostGIS is a separate package from PostgreSQL. On Debian/Ubuntu:

```
sudo apt install -y postgresql-16-postgis-3
```

2 — Enable PostGIS in the database

Connect to the `portsmith` database and enable the extension:

```
psql portsmith
```

```
CREATE EXTENSION IF NOT EXISTS postgis;
```

Verify it loaded:

```sql
SELECT postgis_full_version();
```

You should see a long string beginning with `POSTGIS="3.x.x"`. If you see an error about the extension not existing, the server package is not installed.

Loading the Data

Prerequisites

Chapter 1's seed script must have been run first — the `businesses` table must exist. If it does not:

```
python data/ch01_seed.py
```

Run the Chapter 2 seed

```
python data/ch02_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Enabling PostGIS extension …
Applying DDL …
Inserting 6 neighbourhoods …
Inserting 6 parks …
Inserting 12 road segments …
Updating 48 business locations …

Done:
  businesses with geometry : 48
  neighbourhoods           : 6
  parks                    : 6
  road segments            : 12
```

Verify the load

Open `psql portsmith` and run these four checks.

Check 1 — businesses now has a geometry column:

```
\d businesses
```

You should see a new `geom` column of type `geometry(Point,4326)`.

Check 2 — neighbourhood table structure and row count:

```
SELECT name, population,
       ST_AsText(geom) AS wkt_preview
FROM   neighborhoods
ORDER  BY name;
```

```
       name         | population |            wkt_preview
--------------------+------------
+-----------------------------------
 Harbour District   |       4200 | POLYGON((-1.805 50.69,...))
 Industrial Port    |       2100 | POLYGON((-1.77 50.69,...))
 Northgate          |      18500 | POLYGON((-1.83 50.732,...))
 Old Town           |       6800 | POLYGON((-1.805 50.71,...))
 Riverside          |       9300 | POLYGON((-1.773 50.71,...))
 University Quarter |      11200 | POLYGON((-1.83 50.71,...))
(6 rows)
```

Check 3 — parks and roads:

```
SELECT COUNT(*) FROM parks;
SELECT COUNT(*) FROM city_infrastructure;
```

Both should return 6 and 12 respectively.

Check 4 — SRID on all tables:

```
SELECT f_table_name, f_geometry_column, srid, type
FROM   geometry_columns
WHERE  f_table_schema = 'public'
ORDER  BY f_table_name;
```

```
      f_table_name       | f_geometry_column | srid |    type
-------------------------+-------------------+------+------------
 businesses              | geom              | 4326 | POINT
 city_infrastructure     | geom              | 4326 | LINESTRING
 neighborhoods           | geom              | 4326 | POLYGON
 parks                   | geom              | 4326 | POLYGON
(4 rows)
```

If all four pass, proceed to the exercises.

Exercises

Exercise 1 — Neighbourhood Polygons and WKT

Geometry in PostGIS is often loaded from **Well-Known Text** (WKT), a human-readable representation of shapes. Understanding WKT lets you read geometry from SQL output, write it in queries, and reason about what you stored.

1.1 — Read a polygon in WKT

```
SELECT name, ST_AsText(geom) AS wkt
FROM   neighborhoods
WHERE  name = 'Harbour District';
```

```
       name        |                          wkt
-------------------
+----------------------------------------------------
 Harbour District  | POLYGON((-1.805 50.69,-1.77 50.69,-1.77
50.71,...))
```

WKT for a polygon is `POLYGON((x1 y1, x2 y2, ...))`. In geographic coordinates the convention is *longitude latitude* (x then y), matching the X/Y convention in maths. The ring must **close** — the last coordinate must equal the first.

1.2 — Inspect the bounding box

`ST_Envelope` returns the minimum bounding rectangle for any geometry:

```
SELECT name,
       ST_XMin(ST_Envelope(geom)) AS west,
       ST_XMax(ST_Envelope(geom)) AS east,
       ST_YMin(ST_Envelope(geom)) AS south,
       ST_YMax(ST_Envelope(geom)) AS north
FROM   neighborhoods
ORDER  BY name;
```

Use this output to confirm each polygon sits in the right part of the coordinate space (longitudes around −1.75 to −1.83, latitudes around 50.69 to 50.76).

1.3 — Count vertices

```
SELECT name,
       ST_NPoints(geom) AS vertex_count
FROM   neighborhoods
ORDER  BY name;
```

Each neighbourhood polygon has five vertices (four corners plus the closing duplicate). A real city boundary imported from an OS/census shapefile might have thousands.

1.4 — Understand the SRID

`ST_SRID` returns the spatial reference identifier stored with the geometry:

```
SELECT name, ST_SRID(geom) AS srid
FROM   neighborhoods
LIMIT  3;
```

SRID 4326 is the WGS-84 system used by GPS. Every geometry in these tables carries the same SRID, which means they can be compared and joined directly. If SRIDs differ, PostGIS will return an error — a deliberate safety check.

> **The geometry/geography distinction:** In PostGIS, `geometry` stores coordinates in whatever units the SRS defines. For SRID 4326 that means degrees. When you ask for a distance between two `geometry` points in 4326, you get a value in *degrees* — nearly useless for human-scale distances. The `geography` type, by contrast, always works on a spheroid and returns distances in *metres*. You can cast any SRID-4326 geometry to geography with `::geography` to get metre-based calculations. The exercises use this cast throughout.

Exercise 2 — Proximity Search with `ST_DWithin`

`ST_DWithin(a, b, distance)` returns true when the distance between `a` and `b` is at most `distance`. With geography inputs the distance is in metres.

2.1 — Find businesses within 500 m of Portsmith Pier

Portsmith's main pier entrance sits at approximately (−1.785, 50.700). Find every business within 500 metres of it:

```
SELECT b.name,
       b.neighbourhood,
       ROUND(
           ST_Distance(b.geom::geography,
                       ST_Point(-1.785,
50.700)::geography)::numeric
       ) AS distance_m
FROM   businesses b
```

```sql
WHERE  ST_DWithin(b.geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500)
ORDER  BY distance_m;
```

```
          name           |  neighbourhood   | distance_m
-------------------------+-----------------+------------
 The Gilded Clam         | Harbour District |         70
 Anchor & Oar Tavern     | Harbour District |        179
 Tidal Wave Surf Shop    | Harbour District |        307
 Harbour Inn             | Harbour District |        437
 Portsmith Fish Market   | Harbour District |        484
(5 rows)
```

All five results are in the Harbour District — exactly what you would expect for a search centred on the pier.

2.2 — Why `::geography` matters

Try the same query without the cast. Note that `ST_Point(...)` without an explicit SRID gets SRID 0, which PostGIS refuses to compare against SRID-4326 geometry — so you must use `ST_SetSRID`:

```sql
SELECT b.name,
       ROUND(ST_Distance(b.geom,
                         ST_SetSRID(ST_Point(-1.785, 50.700),
4326))::numeric,
             6)                AS distance_degrees
FROM   businesses b
WHERE  ST_DWithin(b.geom,
                  ST_SetSRID(ST_Point(-1.785, 50.700), 4326),
                  500)
ORDER  BY distance_degrees;
```

The `WHERE` clause now has a threshold of `500` — but in *degrees*. One degree of latitude is about 111 km, so this matches businesses within roughly 55,000 km: all 48 of them, from the whole city. The `distance_degrees` values in the output are tiny fractions like `0.001000`, not useful numbers.

```
rows returned: 48  (the entire city, not 5)
```

> **Rule:** For any distance query on SRID-4326 data, always cast to `::geography` in `ST_DWithin` and `ST_Distance`. The cast has negligible performance cost at the scales used in city-level data.

2.3 — Try different radii

How many businesses fall within 1 km of the pier? 2 km?

```sql
SELECT radius_m,
       COUNT(*) AS business_count
FROM   (VALUES (500), (1000), (2000)) AS radii(radius_m)
CROSS JOIN LATERAL (
    SELECT 1
    FROM   businesses b
    WHERE  ST_DWithin(b.geom::geography,
                      ST_Point(-1.785, 50.700)::geography,
                      radius_m)
) AS matches
GROUP  BY radius_m
ORDER  BY radius_m;
```

```
 radius_m | business_count
----------+----------------
      500 |              5
     1000 |              8
     2000 |             17
(3 rows)
```

The pier sits in the southern Harbour District. Doubling the radius to 2 km pulls in more of the central neighbourhoods, but the northern Northgate businesses are nearly 5 km away — a reminder that Portsmith stretches a significant distance north from the waterfront.

Exercise 3 — Spatial Joins with `ST_Within` and `ST_Contains`

A **spatial join** links rows from two tables based on a geometric relationship rather than a key match. The most common are containment tests: does point A fall inside polygon B?

3.1 — Which neighbourhood is each business in?

```sql
SELECT b.name,
       b.neighbourhood          AS declared_neighbourhood,
       n.name                   AS postgis_neighbourhood
FROM   businesses  b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)
ORDER  BY n.name, b.name;
```

`ST_Within(a, b)` returns true when geometry `a` lies completely inside geometry `b`. The query should return all 48 businesses, each matched to its neighbourhood.

Confirm the declared `neighbourhood` column always matches `postgis_neighbourhood`:

```sql
SELECT COUNT(*) AS mismatches
FROM   businesses  b
```

```
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)
WHERE  b.neighbourhood <> n.name;
```

```
 mismatches
------------
          0
(1 row)
```

Zero mismatches: the text column from Chapter 1 and the geometry are consistent.

3.2 — `ST_Contains` is the inverse

`ST_Contains(a, b)` returns true when polygon `a` contains geometry `b` — it is the mirror of `ST_Within`. These two queries produce identical results:

```sql
-- Point inside polygon
SELECT b.name FROM businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom)
WHERE  n.name = 'Old Town'
ORDER  BY b.name;

-- Polygon contains point
SELECT b.name FROM businesses b
JOIN   neighborhoods n ON ST_Contains(n.geom, b.geom)
WHERE  n.name = 'Old Town'
ORDER  BY b.name;
```

```
          name
--------------------------
 Bella Napoli
 Finch & Sons Barbers
 Le Petit Bistro
 Old Town Hardware
 Portsmith Accountancy Ltd.
 Portsmith Arms Hotel
 Portsmith Legal Group
 Portsmith Tailors
 The Clocktower Pub
(9 rows)
```

Both return the same 9 Old Town businesses.

> **The subtle difference:** `ST_Contains(A, B)` requires that no point of B lies on the boundary of A. A point sitting exactly *on* a polygon edge would fail `ST_Contains` but pass `ST_Covers`. In practice, coordinates rarely land precisely on a boundary, so the

> two functions behave identically for most point-in-polygon work. Reach for `ST_Covers` if you need to include boundary-touching cases explicitly.

3.3 — Check for gaps: businesses in no neighbourhood

This query finds any businesses whose geometry falls outside every neighbourhood polygon — useful for catching data quality problems:

```
SELECT b.name, b.neighbourhood
FROM   businesses b
WHERE  NOT EXISTS (
    SELECT 1
    FROM   neighborhoods n
    WHERE  ST_Within(b.geom, n.geom)
);
```

```
 name | neighbourhood
------+---------------
(0 rows)
```

All 48 businesses are inside a neighbourhood polygon.

Exercise 4 — Computing Area in Square Kilometres

`ST_Area` returns the area of a polygon. Called on `geometry` (degrees), it returns a value in square degrees — meaningless to most people. Called on `geography`, it returns square metres.

4.1 — Incorrect: area in square degrees

```
SELECT name,
       ROUND(ST_Area(geom)::numeric, 6) AS area_sq_degrees
FROM   neighborhoods
ORDER  BY area_sq_degrees DESC;
```

The numbers look tiny (around `0.0007`). They are geometrically correct but impossible to interpret as real-world area because degrees are not uniform units.

4.2 — Correct: area in square metres, then kilometres

```
SELECT name,
       ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS
area_km2
FROM   neighborhoods
ORDER  BY area_km2 DESC;
```

```
       name         | area_km2
--------------------+----------
 Northgate          |    17.59
 Old Town           |     5.53
 Harbour District   |     5.50
 University Quarter |     4.32
 Riverside          |     3.98
 Industrial Port    |     3.14
(6 rows)
```

Northgate is by far the largest neighbourhood — it spans the entire northern width of the city. Industrial Port is the smallest, a tight strip of dockside land. Old Town and Harbour District are almost identical in area despite having very different characters.

> **Why the numbers differ slightly from manual estimates:** `ST_Area` on geography uses the WGS-84 spheroid, which accounts for the fact that the Earth is not a perfect sphere. At latitude 50.7° the correction is small (under 0.3%) but present.

4.3 — Population density

Combine the computed area with the `population` column:

```
        SELECT name,
               population,
               ROUND((ST_Area(geom::geography) / 1e6)::numeric, 2) AS
area_km2,
               ROUND(
                   (population / (ST_Area(geom::geography) /
1e6))::numeric
               ) AS pop_per_km2
        FROM   neighborhoods
        ORDER  BY pop_per_km2 DESC;
```

```
       name         | population | area_km2 | pop_per_km2
--------------------+------------+----------+-------------
 University Quarter |      11200 |     4.32 |        2592
 Riverside          |       9300 |     3.98 |        2340
 Old Town           |       6800 |     5.53 |        1230
 Northgate          |      18500 |    17.59 |        1052
 Harbour District   |       4200 |     5.50 |         763
 Industrial Port    |       2100 |     3.14 |         668
(6 rows)
```

The University Quarter and Riverside are the most densely populated neighbourhoods — student housing and riverside apartments pack a lot of people into compact areas. The Industrial Port is the least dense despite its small size; much of it is warehouse and dock rather than residential.

Exercise 5 — Nearest Park Using `ST_Distance` and a Lateral Join

Finding the nearest feature from another table requires a **lateral join** — a subquery that can reference columns from the outer query row by row.

5.1 — Distance from one business to one park

`ST_Distance` between two geography values returns metres:

```
        SELECT b.name                             AS business,
               p.name                             AS park,
               ROUND(ST_Distance(b.geom::geography,
                                 p.geom::geography)::numeric) AS
distance_m
        FROM   businesses b,
               parks p
        WHERE  b.name = 'The Gilded Clam'
        ORDER  BY distance_m;
```

```
     business     |            park            | distance_m
-----------------+---------------------------+------------
 The Gilded Clam  | Harbourside Park          |        682
 The Gilded Clam  | Dockside Green            |       1315
 The Gilded Clam  | Market Square Gardens     |       2143
 The Gilded Clam  | Riverside Walk Park       |       2899
 The Gilded Clam  | University Grounds        |       3060
 The Gilded Clam  | Northgate Recreation Ground |       5006
(6 rows)
```

The nearest park to The Gilded Clam is Harbourside Park, 682 m away. The `ST_Distance` to a polygon returns the distance from the point to the nearest point on the polygon's boundary — zero when the point is inside the polygon.

5.2 — Nearest park for a single business (lateral pattern)

The canonical pattern for "nearest one thing" is `ORDER BY ... LIMIT 1` inside a lateral subquery:

```
        SELECT b.name,
               nearest.park_name,
               nearest.distance_m
        FROM   businesses b
        CROSS JOIN LATERAL (
            SELECT p.name                                              AS
park_name,
                   ROUND(ST_Distance(b.geom::geography,
```

```
                                  p.geom::geography)::numeric)      AS
distance_m
            FROM   parks p
            ORDER  BY b.geom::geography <-> p.geom::geography
            LIMIT  1
        ) AS nearest
        WHERE  b.name = 'Quarter Note Jazz Club';

         name          |     park_name       | distance_m
-----------------------+---------------------+------------
 Quarter Note Jazz Club | University Grounds |        233
(1 row)
```

The <-> operator is the KNN (k-nearest-neighbour) distance operator. When used in an `ORDER BY` clause inside a lateral join, PostGIS can accelerate it with the GIST index (which you will create in Exercise 6). Using <-> in `ORDER BY` is preferred over `ORDER BY ST_Distance(...)` for this reason.

5.3 — Nearest park for every business

Remove the `WHERE` filter to run across all 48 businesses:

```
        SELECT b.name                          AS business,
               b.neighbourhood,
               nearest.park_name,
               nearest.distance_m
        FROM   businesses b
        CROSS JOIN LATERAL (
            SELECT p.name                                          AS
park_name,
                   ROUND(ST_Distance(b.geom::geography,
                                  p.geom::geography)::numeric)      AS
distance_m
            FROM   parks p
            ORDER  BY b.geom::geography <-> p.geom::geography
            LIMIT  1
        ) AS nearest
        ORDER  BY b.neighbourhood, b.name;
```

Scan the results. You should see a clean pattern: each neighbourhood's businesses point to the park in that same neighbourhood. Notice that three Riverside businesses (Portsmith Pharmacy, Portsmith Veterinary Clinic, Riverside Cinema) and University Bookshop all show `distance_m = 0` — their coordinates fall *inside* the park polygon. `ST_Distance` to a polygon returns zero when the point is contained within it.

5.4 — Aggregate: average walking distance to a park, by neighbourhood

```sql
SELECT b.neighbourhood,
       ROUND(AVG(nearest.distance_m)::numeric) AS avg_distance_m
FROM   businesses b
CROSS JOIN LATERAL (
    SELECT ROUND(ST_Distance(b.geom::geography,
                             p.geom::geography)::numeric) AS
distance_m
    FROM   parks p
    ORDER  BY b.geom::geography <-> p.geom::geography
    LIMIT  1
) AS nearest
GROUP  BY b.neighbourhood
ORDER  BY avg_distance_m;
```

This gives a rough "walkability to green space" metric per neighbourhood. Old Town ranks first — Market Square Gardens sits centrally within the neighbourhood. Northgate comes last despite having the largest park, because the park is in the far north of the district while businesses cluster near the southern edge.

Exercise 6 — GIST Indexes and Spatial Query Plans

Without an index, every spatial query is a sequential scan: PostgreSQL reads every row, applies the geometry function, and discards non-matching rows. For a 48-row dataset that is instant, but at 48 million rows it is not.

PostGIS spatial queries are accelerated by **GIST** (Generalised Search Tree) indexes. A GIST index on a geometry column builds a tree of bounding boxes, allowing the planner to quickly eliminate large portions of the table.

6.1 — Observe the plan before indexing

```sql
EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500);
```

On the 48-row `businesses` table you will see something like:

```
Seq Scan on businesses  (cost=0.00..4.60 rows=1 width=...) ...
  Filter: (st_dwithin(...))
  Rows Removed by Filter: 43
```

A sequential scan is expected on a tiny table — the planner knows the overhead of an index lookup exceeds the cost of reading all 48 rows.

6.2 — Create GIST indexes on the geometry columns

```
CREATE INDEX idx_businesses_geom
    ON businesses USING GIST (geom);

CREATE INDEX idx_neighborhoods_geom
    ON neighborhoods USING GIST (geom);

CREATE INDEX idx_parks_geom
    ON parks USING GIST (geom);

CREATE INDEX idx_city_infrastructure_geom
    ON city_infrastructure USING GIST (geom);
```

6.3 — The geometry/geography index split

With `enable_seqscan` off, verify which plan the proximity query uses:

```
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500);

SET enable_seqscan = on;
```

You will see — perhaps surprisingly — that the planner *still* uses a sequential scan, even though `idx_businesses_geom` exists:

```
Seq Scan on businesses  (cost=10000000000.00...) ...
  Filter: st_dwithin((geom)::geography, ...)
```

Why? The GIST index was built on `geom` (type `geometry`). The query filters on `geom::geography` (type `geography`). These are different types — the index is not usable for geography operations.

To accelerate geography-based distance queries you need a **functional index** on the cast:

```
CREATE INDEX idx_businesses_geom_geography
    ON businesses USING GIST (CAST(geom AS geography));
```

Now with `enable_seqscan` off:

```sql
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name
FROM   businesses
WHERE  ST_DWithin(geom::geography,
                  ST_Point(-1.785, 50.700)::geography,
                  500);

SET enable_seqscan = on;
```

```
Index Scan using idx_businesses_geom_geography on businesses
  Index Cond: ((geom)::geography &&
             _st_expand('...', '500'))
  Filter: st_dwithin((geom)::geography, ...)
  Rows Removed by Filter: 2
```

The index is now used. The index condition uses `&&` (bounding-box overlap on the spheroid) to quickly discard most rows, then `st_dwithin` rechecks the exact distance for the survivors.

6.4 — Verify the spatial join plan

The containment join uses geometry-to-geometry operations, so `idx_businesses_geom` (the plain geometry index) *is* used there:

```sql
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT b.name, n.name AS neighbourhood
FROM   businesses b
JOIN   neighborhoods n ON ST_Within(b.geom, n.geom);

SET enable_seqscan = on;
```

```
Nested Loop ...
  ->  Seq Scan on neighborhoods n  (6 rows — tiny table)
  ->  Index Scan using idx_businesses_geom on businesses b
        Index Cond: (geom @ n.geom)
        Filter: st_within(geom, n.geom)
```

For each neighbourhood polygon, `idx_businesses_geom` is used to find businesses whose bounding box falls inside the neighbourhood's bounding box (`geom @ n.geom`), then `st_within` rechecks the exact containment.

> **The index rule:** A GIST index on `geom geometry` accelerates geometry-to-geometry operations (`ST_Within`, `ST_Contains`, `ST_Intersects`, `&&`). A GIST index on `CAST(geom AS geography)` accelerates geography operations

> (`ST_DWithin(...::geography, ...)`, `<->` on geography). If you query both ways, create both indexes — they coexist happily on the same table.

Summary — What You Should Now Know

You have worked through the core PostGIS toolkit for points, polygons, and linear features. Here is a reference for everything used:

Function / operator	What it does
`ST_GeomFromText(wkt, srid)`	Parse WKT into a geometry with the given SRID
`ST_AsText(geom)`	Format geometry as WKT for display
`ST_SetSRID(ST_MakePoint(lon, lat), srid)`	Construct a point geometry
`ST_Point(lon, lat)::geography`	Construct a point and cast to geography
`geom::geography`	Cast a 4326 geometry to geography (distances now in metres)
`ST_SRID(geom)`	Return the SRID stored with a geometry
`ST_NPoints(geom)`	Count vertices in a geometry
`ST_Envelope(geom)`	Return the bounding-box rectangle
`ST_XMin/XMax/YMin/YMax(geom)`	Extract bounding-box extents
`ST_DWithin(a, b, d)`	True when the distance between a and b is ≤ d
`ST_Distance(a, b)`	Distance between two geometries (metres for geography)
`a::geography <-> b::geography`	KNN distance operator; use in ORDER BY for index acceleration
`ST_Within(a, b)`	

Function / operator	What it does
	True when a lies completely inside b
`ST_Contains(a, b)`	True when a contains b (inverse of ST_Within)
`ST_Area(geom::geography)`	Area in square metres on the spheroid
`CROSS JOIN LATERAL (... LIMIT 1)`	Nearest-neighbour join pattern
`USING GIST (geom)`	Create a GIST spatial index

> **Geometry vs geography in one sentence:** Use `geometry` for storage and when working with projected coordinate systems where units are already metres or feet. Cast to `geography` any time you need distance, area, or proximity results in real-world units from SRID-4326 data.

The coordinates added in this chapter will carry forward. Chapter 12 extends the `city_infrastructure` road network into a graph and uses a recursive CTE to find the shortest path between two intersections.

Going further: PostGIS supports many more geometry types — `MULTIPOLYGON` for areas with holes, `GEOMETRYCOLLECTION` for mixed types, and 3D geometries with a Z coordinate for elevation data. For routing specifically, `pgRouting` builds on PostGIS to provide Dijkstra, A, and turn-restriction-aware shortest-path algorithms over road network graphs. For importing real boundary data, `shp2pgsql` converts ESRI Shapefiles directly into PostGIS-compatible `INSERT` statements, and `ogr2ogr` handles GeoJSON, KML, GeoPackage, and dozens of other formats.*

Chapter 3 — Job Queues: `FOR UPDATE SKIP LOCKED`

> *"A queue is just a table that everyone is racing to read."*

Background

Sooner or later almost every application needs a queue: a list of work items that a pool of workers processes one at a time, safely, without two workers ever grabbing the same item. The reflexive answer is to reach for a message broker — Redis, RabbitMQ, SQS, Kafka. Those tools earn their keep at serious scale. But if your data already lives in PostgreSQL, running a second system just to hand out rows to workers is often unnecessary complexity: another service to deploy, monitor, and keep consistent with the database.

PostgreSQL can do this job itself. The `FOR UPDATE SKIP LOCKED` row-locking clause, combined with an ordinary table, gives you an atomic "claim the next item and don't let anyone else touch it" primitive — the same guarantee a dedicated queue product sells you, built out of two SQL keywords. This chapter builds a job queue from scratch: the schema, the atomic claim query, concurrent worker behaviour, stalled-job recovery, and a dead-letter path for jobs that keep failing.

This is not a toy exercise. This exact pattern — a status column, a claim query, a heartbeat, a dead-letter table — is what libraries like `river`, `oban` (Elixir), and countless in-house job runners implement on top of PostgreSQL in production.

The Scenario

Portsmith's permitting office processes a steady stream of permit applications: building work, business licenses, public events, signage, and demolitions. Each application needs to move through a review pipeline, and the office wants that processing to happen asynchronously and reliably — work should never be lost, never double-processed, and a crashed reviewer process shouldn't leave an application stuck in limbo forever.

The `jobs` table models this as a queue. Every row is one permit application awaiting review. A `status` column tracks its life cycle, a `priority` column lets safety-critical work (demolitions) jump ahead of routine work (sign permits), and a retry counter with a companion `dead_letter_jobs` table handles applications whose processing keeps failing.

Column	Purpose
`status`	`queued` → `in_progress` → `completed` (or back to `queued`, or dead-lettered)
`priority`	1 (most urgent — demolitions) to 5 (least urgent — sign permits)
`payload`	JSONB — the permit application details
`attempts` / `max_attempts`	Retry bookkeeping
`claimed_by` / `claimed_at`	Which worker has the job, and since when
`heartbeat_at`	Updated periodically by the worker while it holds the job

No extensions are required for this chapter — everything here is built on core PostgreSQL locking semantics.

Exercise Goals

By the end of this chapter you will be able to:

- Design a queue table schema, including a partial index tuned for the claim query's access pattern.
- Write the atomic `UPDATE ... FROM (SELECT ... FOR UPDATE SKIP LOCKED)` claim query, and explain why naive two-step approaches race.
- Observe, with two concurrent `psql` sessions, that `SKIP LOCKED` lets workers proceed past rows another worker is holding — and contrast that with the blocking behaviour of plain `FOR UPDATE`.
- Implement a heartbeat/timeout mechanism that reclaims jobs abandoned by a crashed worker.
- Route jobs that exhaust their retries into a dead-letter table.

- Benchmark claim throughput at different concurrency levels with `pgbench`.

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. You will also use `pgbench` for Exercise 6, which ships with the standard PostgreSQL client tools (`postgresql-client-16` on Debian/Ubuntu).

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch03_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema …
Inserting 45 jobs …
Done — 45 rows in jobs, all queued.
```

The seed script is self-contained — it does not depend on Chapter 1 or 2's data.

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d jobs
```

```
                                        Table "public.jobs"
    Column    |           Type           | Collation | Nullable
|             Default
--------------+--------------------------+-----------+----------
+---------------------------------
 id           | bigint                   |           | not null |
nextval('jobs_id_seq'::regclass)
```

```
 job_type     | text                     |          | not null |
 payload      | jsonb                    |          | not null |
 status       | text                     |          | not null |
'queued'::text
 priority     | smallint                 |          | not null |
5
 attempts     | integer                  |          | not null |
0
 max_attempts | integer                  |          | not null |
3
 created_at   | timestamp with time zone |          | not null |
clock_timestamp()
 claimed_at   | timestamp with time zone |          |          |
 claimed_by   | text                     |          |          |
 heartbeat_at | timestamp with time zone |          |          |
 completed_at | timestamp with time zone |          |          |
 last_error   | text                     |          |          |
Indexes:
    "jobs_pkey" PRIMARY KEY, btree (id)
    "idx_jobs_claim_order" btree (priority, created_at, id) WHERE
status = 'queued'::text
    "idx_jobs_status" btree (status)
Check constraints:
    "jobs_status_check" CHECK (status = ANY (ARRAY['queued'::text,
'in_progress'::text, 'completed'::text, 'failed'::text]))
```

Check 2 — job counts by type and priority:

```sql
SELECT job_type, priority, COUNT(*) AS jobs
FROM   jobs
GROUP  BY job_type, priority
ORDER  BY priority;
```

```
     job_type      | priority | jobs
-------------------+----------+------
 demolition_permit |        1 |    4
 business_license  |        2 |   12
 building_permit   |        3 |   15
 event_permit      |        4 |    8
 sign_permit       |        5 |    6
(5 rows)
```

Check 3 — everything starts queued:

```sql
SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

```
 status | count
--------+-------
 queued |    45
(1 row)
```

If all three match, proceed to the exercises.

> **Note:** If you re-run `ch03_seed.py` at any point during the exercises to reset to a clean state, it drops and recreates both `jobs` and `dead_letter_jobs`.

Exercises

Exercise 1 — Designing the Queue Schema

1.1 — Why a partial index

The claim query (which you'll write in Exercise 2) only ever looks at rows where `status = 'queued'`, ordered by `priority` then `created_at`. As the queue runs, the vast majority of rows will end up `completed` — a normal B-tree index on `(priority, created_at, id)` would faithfully index every one of those settled rows even though the claim query never looks at them.

`idx_jobs_claim_order` is a **partial index** — it only indexes rows matching `WHERE status = 'queued'`:

```
CREATE INDEX idx_jobs_claim_order
    ON jobs (priority, created_at, id)
    WHERE status = 'queued';
```

This keeps the index small regardless of how many historical jobs pile up in `completed` or `failed` state, because settled rows are never in it.

1.2 — Why `id` is part of the sort key, not just `created_at`

You might expect `ORDER BY priority, created_at` to be enough — oldest job in the highest-priority bucket goes first. But timestamps are not always unique. If two jobs are inserted in the same transaction, both can get an identical `created_at` (more on this below), and `ORDER BY` over tied values has no defined order. Appending the primary key, `id`, as a final tiebreaker guarantees a deterministic order even when timestamps collide:

```
ORDER BY priority ASC, created_at ASC, id ASC
```

> **A gotcha worth knowing:** `now()` returns the **transaction's** start time, not the current wall-clock time — every call to `now()` inside the same transaction returns the same value. The seed script originally used `created_at TIMESTAMPTZ DEFAULT now()`, and because all 45 rows were inserted in one transaction, every

> single row ended up with an *identical* `created_at`. The fix is `clock_timestamp()`, which returns the actual current time at the moment it's evaluated, differing row to row even within one transaction. `jobs.created_at` uses `clock_timestamp()` for exactly this reason. This is the same family of surprise as the JSONB `null` vs. SQL `NULL` gotcha from Chapter 1: a function name that looks interchangeable with another is not.

1.3 — Confirm the index is used

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM   jobs
WHERE  status = 'queued'
ORDER  BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT  1;
```

```
 Limit  (cost=0.14..6.17 rows=1 width=24) (actual
time=0.022..0.022 rows=1 loops=1)
   Buffers: shared hit=3
   ->  LockRows  (cost=0.14..12.20 rows=2 width=24) (actual
time=0.021..0.021 rows=1 loops=1)
         Buffers: shared hit=3
         ->  Index Scan using idx_jobs_claim_order on jobs
(cost=0.14..12.18 rows=2 width=24) (actual time=0.010..0.010
rows=1 loops=1)
               Filter: (status = 'queued'::text)
               Buffers: shared hit=2
```

Now drop the index and run the identical query again inside a transaction you roll back (so the drop doesn't stick):

```
BEGIN;
DROP INDEX idx_jobs_claim_order;

EXPLAIN (ANALYZE, BUFFERS)
SELECT id
FROM   jobs
WHERE  status = 'queued'
ORDER  BY priority ASC, created_at ASC, id ASC
FOR UPDATE SKIP LOCKED
LIMIT  1;

ROLLBACK;
```

```
 Limit  (cost=9.51..9.53 rows=1 width=24) (actual
time=0.071..0.072 rows=1 loops=1)
   ->  LockRows  (cost=9.51..9.54 rows=2 width=24) (actual
time=0.070..0.071 rows=1 loops=1)
         ->  Sort  (cost=9.51..9.52 rows=2 width=24) (actual
```

```
time=0.067..0.068 rows=1 loops=1)
               Sort Key: priority, created_at, id
               Sort Method: quicksort  Memory: 27kB
               ->  Bitmap Heap Scan on jobs  (cost=4.16..9.50
rows=2 width=24) (actual time=0.018..0.027 rows=45 loops=1)
                     Recheck Cond: (status = 'queued'::text)
                     ->  Bitmap Index Scan on idx_jobs_status
(cost=0.00..4.16 rows=2 width=0) (actual time=0.013..0.013 rows=45
loops=1)
                           Index Cond: (status = 'queued'::text)
```

Without `idx_jobs_claim_order`, PostgreSQL still finds the queued rows (via `idx_jobs_status`), but it must then **sort all of them** to find the one with the lowest `(priority, created_at, id)` — an extra `Sort` node. With the tailored partial index, the rows are already stored in claim order, so PostgreSQL walks the index and stops at the first match. On a 45-row table the difference is invisible; on a busy production queue with a deep backlog, eliminating the sort on every single claim matters a great deal.

Exercise 2 — The Atomic Claim Query

2.1 — Why a naive two-step claim races

The tempting first approach is to `SELECT` a candidate row, then `UPDATE` it in a second statement:

```sql
        --   Do not do this — it has a race condition
        SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority,
created_at, id LIMIT 1;
        -- ... application reads id = 7 ...
        UPDATE jobs SET status = 'in_progress' WHERE id = 7;
```

Between the `SELECT` and the `UPDATE`, nothing stops a second worker from running the exact same `SELECT`, reading the same `id = 7`, and also issuing the `UPDATE`. Both workers now believe they own job 7. This is a classic **check-then-act race condition** — the gap between reading and acting is exactly where two workers can interleave.

2.2 — `FOR UPDATE` closes the gap, but blocks

Locking the row as part of the `SELECT` closes the race:

```sql
        SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority,
created_at, id
        FOR UPDATE LIMIT 1;
```

Now a second worker running the same query, in a separate transaction, **blocks** until the first worker's transaction commits or rolls back — correct, but it means every worker but one sits idle waiting for a lock instead of moving on to a different job. Section 3.1 demonstrates this.

2.3 — `SKIP LOCKED` lets workers move past each other

Adding `SKIP LOCKED` tells PostgreSQL: if the next candidate row is already locked by another transaction, don't wait for it — skip it and consider the row after it.

```
        SELECT id FROM jobs WHERE status = 'queued' ORDER BY priority,
created_at, id
        FOR UPDATE SKIP LOCKED LIMIT 1;
```

This is the piece that makes PostgreSQL usable as a concurrent queue: N workers can run this query at the same instant and each will walk away with a *different* row, with no blocking and no double-claims.

2.4 — The full atomic claim: lock, update, and return in one statement

Locking the row is only half the job — you still need to mark it `in_progress` before releasing the lock, and you want the whole thing to happen as a single round trip. Combine the `SELECT ... FOR UPDATE SKIP LOCKED` with the `UPDATE` using a CTE:

```
        WITH next_job AS (
            SELECT id
            FROM   jobs
            WHERE  status = 'queued'
            ORDER  BY priority ASC, created_at ASC, id ASC
            FOR UPDATE SKIP LOCKED
            LIMIT  1
        )
        UPDATE jobs
        SET    status       = 'in_progress',
               claimed_at   = now(),
               claimed_by   = 'demo-worker',
               heartbeat_at = now(),
               attempts     = attempts + 1
        FROM   next_job
        WHERE  jobs.id = next_job.id
        RETURNING jobs.id, jobs.job_type, jobs.payload ->>
'application_id' AS application_id,
                  jobs.priority, jobs.attempts, jobs.status;
```

```
 id |      job_type      | application_id | priority | attempts |
status
```

```
----+-------------------+----------------+----------+----------
+-------------
  1 | demolition_permit | DP-2024-0001   |        1 |        1 |
in_progress
(1 row)
```

This single statement is the entire claim operation: find the best candidate, skip anything locked, mark it claimed, and hand back its data — atomically, with no window for a race. This is exactly the query `data/ch03_worker.py` runs (see `CLAIM_SQL`). Run it again and the demolition permit at `id = 2` comes back next — `id = 1` is now `in_progress`, so it's no longer a candidate.

> Roll this back if you ran it directly in `psql` and want to keep the queue clean for later exercises: wrap it in `BEGIN; ... ROLLBACK;`.

Exercise 3 — Simulating Concurrent Workers

3.1 — `SKIP LOCKED` vs. plain `FOR UPDATE`, in two `psql` sessions

Open two terminals with `psql portsmith` in each. In **Session A**, start a transaction, claim 5 rows, and hold the transaction open (don't commit yet):

```sql
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

```
 id |     job_type
----+-------------------
  1 | demolition_permit
  2 | demolition_permit
  3 | demolition_permit
  4 | demolition_permit
  5 | business_license
(5 rows)
```

Leave that transaction open. In **Session B**, run the identical query:

```sql
-- Session B (Session A is still open, holding locks on ids 1-5)
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE SKIP LOCKED LIMIT 5;
```

```
 id |    job_type
----+------------------
```

```
  6 | business_license
  7 | business_license
  8 | business_license
  9 | business_license
 10 | business_license
(5 rows)
```

Session B returns immediately with a **different** set of rows — it silently skipped 1–5 because they were locked, and moved on to the next five unlocked candidates. Commit or roll back both sessions to release the locks:

```sql
COMMIT;   -- run in both sessions
```

3.2 — Now try it with plain `FOR UPDATE` (no `SKIP LOCKED`)

Repeat the same two-session experiment, but drop `SKIP LOCKED`:

```sql
-- Session A
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
-- (leave this transaction open)

-- Session B — this will hang
BEGIN;
SELECT id, job_type FROM jobs WHERE status = 'queued'
ORDER BY priority, created_at, id
FOR UPDATE LIMIT 5;
```

Session B does not return. It is blocked, waiting for Session A's row locks to be released. Only once Session A runs `COMMIT` (or `ROLLBACK`) does Session B's query complete — and when it does, it returns the *same* five rows Session A had, now that they're unlocked again:

```
-- (Session B, after Session A commits)
 id |     job_type
----+-------------------
  1 | demolition_permit
  2 | demolition_permit
  3 | demolition_permit
  4 | demolition_permit
  5 | business_license
(5 rows)
```

This is the difference in one sentence: **plain `FOR UPDATE` serializes workers through the lock; `SKIP LOCKED` lets them fan out across the table.** For a job queue, you always want the latter — a blocked worker is a wasted worker.

3.3 — Real concurrent workers with `ch03_worker.py`

Reset the data (`python data/ch03_seed.py`) if you ran the manual claim in Exercise 2.4 without rolling it back. Then launch two workers in the background, each capped to 10 jobs so they finish quickly:

```
        python data/ch03_worker.py --worker-id w1 --max-jobs 10 --fail-
rate 0 &
        python data/ch03_worker.py --worker-id w2 --max-jobs 10 --fail-
rate 0 &
        wait
```

Each prints a running log as it claims and completes jobs, e.g.:

```
[w1] claimed job 2 (demolition_permit, attempt 1/3): DP-2024-0002
— processing for 0.09s
[w1] job 2 completed
[w2] claimed job 1 (demolition_permit, attempt 1/3): DP-2024-0001
— processing for 0.12s
[w2] job 1 completed
```

Once both finish, confirm the split was clean — 20 jobs completed total, no job claimed by both workers:

```
        SELECT claimed_by, COUNT(*) FROM jobs WHERE status = 'completed'
        GROUP BY claimed_by ORDER BY claimed_by;
```

```
 claimed_by | count
------------+-------
 w1         |    10
 w2         |    10
(2 rows)
```

```
        SELECT status, COUNT(*) FROM jobs GROUP BY status;
```

```
  status   | count
-----------+-------
 completed |    20
 queued    |    25
(2 rows)
```

10 + 10 = 20, matching exactly the `--max-jobs 10` cap on each worker — no overlaps, no double-processing, no lost jobs.

Exercise 4 — Heartbeat and Stalled-Job Recovery

Claiming a job is only half the reliability story. What happens if the worker that claimed a job crashes, gets OOM-killed, or loses its network connection halfway through? The row is stuck at `status = 'in_progress'` forever unless something notices and puts it back.

4.1 — The heartbeat column

`heartbeat_at` exists for exactly this. A well-behaved worker updates it periodically while it holds a job (see the `while` loop in `process_job()` in `ch03_worker.py`, which sends a heartbeat roughly every 2 seconds during simulated processing). If a job has been `in_progress` for a long time *and* its heartbeat has gone stale, that's a strong signal the worker holding it is dead — a live worker would have updated it.

4.2 — Simulate a crashed worker

Pick any queued job and manually walk it through what a real claim would do, then simulate a crash by never sending a heartbeat again:

```
UPDATE jobs
SET    status = 'in_progress', claimed_at = now(),
       claimed_by = 'worker-crashed', heartbeat_at = now(),
       attempts = attempts + 1
WHERE  id = 21;

-- Simulate 5 minutes of silence from the "crashed" worker
UPDATE jobs SET heartbeat_at = now() - interval '5 minutes' WHERE
id = 21;
```

4.3 — Run the reclaim sweep

```
python data/ch03_reclaim.py --timeout 30
```

```
Connecting to: dbname=portsmith
  job 21 (building_permit) stalled — requeued (attempt 1)

Done: 1 requeued, 0 dead-lettered.
```

The sweep looks for `in_progress` rows whose `heartbeat_at` is older than `--timeout` seconds (`FIND_STALLED_SQL` in `ch03_reclaim.py`), and since this job's `attempts` (1) is still below `max_attempts` (3), it goes back to `queued`:

```
SELECT id, status, attempts, claimed_by, heartbeat_at, last_error
FROM   jobs WHERE id = 21;
```

```
 id | status | attempts | claimed_by | heartbeat_at
|                      last_error
```

```
----+--------+---------+-----------+-------------
+---------------------------------------------------
 21 | queued |       1 |           |             | stalled: no
heartbeat since ... (last claimed by worker-crashed)
```

Notice `attempts` stayed at `1` — the reclaim did not reset it. The attempt the crashed worker used is still counted; the job doesn't get a free retry just because the worker that lost it never got to report failure honestly.

4.4 — Why the timeout, not an outright deadline

A fixed timeout on the *last heartbeat* (rather than on total processing time) lets jobs run arbitrarily long as long as they keep proving they're alive. A 10-minute report-generation job with a 30-second heartbeat interval will never be mistaken for stalled, while a worker that dies mid-task is detected within one missed heartbeat window. Run `ch03_reclaim.py` again immediately — it correctly finds nothing to do, since the reclaimed job now has a fresh `queued` state and no stale `in_progress` row exists:

```
No jobs stalled beyond 30s — nothing to do.
```

Exercise 5 — Dead-Lettering Exhausted Jobs

Some jobs will never succeed no matter how many times you retry them — a malformed application, a permanently invalid address, a bug that only triggers on one particular payload. Retrying forever wastes worker time and can mask a real problem. Once a job has used its last attempt, it belongs in `dead_letter_jobs`: out of the active queue, but preserved for a human to inspect.

5.1 — The `dead_letter_jobs` schema

```
        \d dead_letter_jobs
```

```
                     Table "public.dead_letter_jobs"
    Column    |           Type           | Collation | Nullable |
Default
--------------+--------------------------+-----------+----------
+---------
 id           | bigint                   |           | not null |
 job_type     | text                     |           | not null |
 payload      | jsonb                    |           | not null |
 priority     | smallint                 |           | not null |
 attempts     | integer                  |           | not null |
 max_attempts | integer                  |           | not null |
 created_at   | timestamp with time zone |           | not null |
 last_error   | text                     |           |          |
```

```
 failed_at    | timestamp with time zone |            | not null |
now()
```

It mirrors `jobs` but adds `failed_at` and drops the in-flight columns (`status`, `claimed_by`, `heartbeat_at`) that no longer mean anything once a job is out of the active queue.

5.2 — Push a job past its retry limit

Continuing with job 21 from Exercise 4 (now back to `queued` with `attempts = 1`), fast-forward it to its last attempt and let it stall again:

```
        UPDATE jobs
        SET    status = 'in_progress', claimed_at = now(),
               claimed_by = 'worker-crashed-2', heartbeat_at = now() -
interval '5 minutes',
               attempts = max_attempts
        WHERE  id = 21;
```

```
        python data/ch03_reclaim.py --timeout 30
```

```
Connecting to: dbname=portsmith
  job 21 (building_permit) exhausted retries — dead-lettered

Done: 0 requeued, 1 dead-lettered.
```

This time `attempts` (3) is no longer less than `max_attempts` (3), so the sweep's `DEAD_LETTER_SQL` runs instead: it `DELETE`s the row from `jobs` and `INSERT`s it into `dead_letter_jobs` in one CTE, so the row is never visible in neither-place or both-places, even under concurrent access.

```
        SELECT id FROM jobs WHERE id = 21;
```

```
 id
----
(0 rows)
```

```
        SELECT id, job_type, payload ->> 'application_id' AS
application_id,
               attempts, max_attempts, last_error, failed_at
        FROM   dead_letter_jobs WHERE id = 21;
```

```
 id |    job_type     | application_id | attempts | max_attempts
|                      last_error                      |
failed_at
----+-----------------+----------------+----------+-------------
+------------------------------------------------------
+-------------------------------
 21 | building_permit | BP-2024-0005   |        3 |            3 |
stalled: no heartbeat since ... (last claimed ...)   | 2026-07-12
22:15:45.613583-04
(1 row)
```

The exact same dead-lettering logic runs inline inside `ch03_worker.py` when a job fails organically (not via a stall) on its last attempt — see `DEAD_LETTER_SQL` in that script. Whether a job dies from an explicit failure or from going silent, it ends up in the same place, with a record of why.

Exercise 6 — Benchmarking Claim Throughput with `pgbench`

6.1 — A benchmark script that doesn't drain the queue

`pgbench` repeatedly runs a SQL script against the database for a fixed duration, from any number of concurrent client connections — exactly the tool for measuring how the claim query holds up under contention. The catch: if the script just claims a job and leaves it claimed, 45 rows disappear from the pool in well under a second and every worker after that finds nothing to do. `data/ch03_claim_bench.sql` works around this by claiming a job and then **immediately releasing it back to `queued`** in the same transaction:

```
BEGIN;

WITH next_job AS (
    SELECT id
    FROM   jobs
    WHERE  status = 'queued'
    ORDER  BY priority ASC, created_at ASC, id ASC
    FOR UPDATE SKIP LOCKED
    LIMIT  1
)
UPDATE jobs
SET    status       = 'in_progress',
       claimed_at   = now(),
       claimed_by   = 'bench-' || :client_id,
       heartbeat_at = now()
FROM   next_job
WHERE  jobs.id = next_job.id
RETURNING jobs.id AS claimed_id \gset

UPDATE jobs
SET    status = 'queued', claimed_at = NULL, claimed_by = NULL,
heartbeat_at = NULL
WHERE  id = :claimed_id;

COMMIT;
```

`:client_id` is a `pgbench` built-in variable — the number of the simulated client running this script. `\gset` captures the claimed row's `id` into a `pgbench` variable (`:claimed_id`) so the release step can target it. This keeps the pool at a constant 45 claimable rows for the whole benchmark run, so throughput reflects lock contention rather than the queue running dry.

6.2 — Run at increasing concurrency

```
pgbench -n -c 1  -j 1  -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 4  -j 4  -T 5 -f data/ch03_claim_bench.sql portsmith
pgbench -n -c 16 -j 16 -T 5 -f data/ch03_claim_bench.sql portsmith
```

`-c` is the number of simulated concurrent workers (clients), `-j` the number of `pgbench` threads driving them, and `-T 5` runs each test for 5 seconds. Results on the machine used to write this chapter:

Clients (-c)	tps	Avg. latency
1	799	1.25 ms
4	1,202	3.33 ms
16	2,570	6.23 ms

Your numbers will differ with your hardware — the shape of the result is what matters. Throughput climbs with concurrency because `SKIP LOCKED` lets additional workers keep finding unlocked rows to claim instead of queueing up behind each other; latency per transaction also climbs because more workers are contending for the same 45-row table and for CPU. Try dropping `idx_jobs_claim_order` (inside a `BEGIN; ... ROLLBACK;` block, as in Exercise 1.3) and rerunning the `-c 16` case — you should see tps fall, since every claim now pays for a `Sort` over all queued rows instead of an index walk.

6.3 — Where this stops being enough

`pgbench` here is exercising claim contention on 45 rows — a stand-in for "the working set the claim query actually scans." A real permitting-office queue might hold a very different shape of data (thousands of queued rows, heavy skew toward one `priority` bucket), and the honest way to benchmark your own workload is to seed a table that resembles it, not a 45-row toy. Chapter 20 (`pg_stat_statements` and Query Performance) picks this thread back up: rather than a synthetic benchmark, it measures and fixes real slow queries observed in production traffic.

Summary — What You Should Now Know

You built a working, concurrency-safe job queue entirely out of core PostgreSQL features. Here is a reference for the pieces:

Tool	What it does
`FOR UPDATE`	Locks selected rows; other transactions selecting the same rows **block** until released
`FOR UPDATE SKIP LOCKED`	Locks selected rows; other transactions **skip** already-locked rows instead of blocking
`WITH next_job AS (... FOR UPDATE SKIP LOCKED LIMIT 1) UPDATE ... FROM next_job`	The atomic claim pattern: find, lock, and mark a row in one round trip
Partial index `WHERE status = 'queued'`	Keeps the claim-path index small regardless of how many settled rows accumulate
`clock_timestamp()` vs. `now()`	`now()` freezes at transaction start; `clock_timestamp()` reflects real wall-clock time on every call
`heartbeat_at` + timeout sweep	Detects and recovers jobs abandoned by a crashed worker
`dead_letter_jobs`	Removes permanently-failing jobs from the active queue while preserving them for inspection
`pgbench -f script.sql`	Benchmarks a custom SQL workload at a chosen concurrency level

The key design insight from this chapter is that a reliable queue is not one clever query — it's the combination of an atomic claim, a way to detect workers that go silent, and a place for work that genuinely cannot succeed. PostgreSQL's row-locking primitives give you the first two almost for free; the dead-letter table is just a normal table.

The `jobs` table you built here is reused directly in later chapters: Chapter 13 adds a trigger that fires `NOTIFY` on status changes so listeners can react to queue activity in real time, Chapter 14 uses advisory locks alongside it for leader-election patterns, and Chapter 19 schedules the reclaim sweep you wrote by hand in Exercise 4 to run automatically with `pg_cron`.

Going further: the pattern in this chapter — status column, atomic claim, heartbeat, dead-letter — is exactly what PostgreSQL-backed job-queue libraries like `river` (Go) and `oban` (Elixir) implement, with more polish around scheduling, uniqueness constraints, and observability. If you outgrow a single-database queue — cross-database work distribution, guaranteed ordering at very high throughput, or consumer groups — that's the point at which a dedicated broker starts to earn its operational cost. For most applications below that scale, the table you just built is enough.

Chapter 4 — Full-Text Search: `tsvector`, Stopwords, and Ranking

> *“Grep finds characters. Full-text search finds meaning — or at least gets a lot closer.”*

Background

`LIKE '%harbour%'` finds the literal substring “harbour”. It will not find “Harbour”, “harbours”, or a document about “the harbor” (different spelling) unless you handle every variation yourself, and it cannot tell you whether a match in the title is more relevant than a match buried in paragraph six. As soon as an application needs to search prose — meeting minutes, support tickets, product descriptions, articles — substring matching stops being enough. The usual reflex is to reach for Elasticsearch or a similar dedicated search engine.

PostgreSQL has had a full-text search engine built in since version 8.3. It tokenizes text into words, reduces those words to normalized root forms (*stemming*), discards low-information words like “the” and “of” (*stopwords*), and stores the result in a specialized `tsvector` type that a GIN index can search in milliseconds. A companion `tsquery` type lets you express boolean and phrase searches, and a family of ranking functions score how well each match fits the query. None of this requires an extension — it is core PostgreSQL, the same engine `websearch_to_tsquery`-powered search boxes on production sites are built on.

This chapter builds a search feature over a small document archive: how text becomes a `tsvector`, what stopword removal actually throws away, how to keep the vector in sync as rows change, how query syntax differs between a developer-facing boolean query language and a plain-text search box, and how to rank and highlight results. The final exercise customizes the stopword list itself — because in a *city* government’s document archive, the words “city” and “council” appear so often they stop being useful signals, exactly the kind of domain-specific tuning full-text search is designed to support.

The Scenario

Portsmith's city clerk publishes an archive of public records: council meeting minutes, zoning ordinances, and public notices. Residents need to search this archive by keyword — "what happened with the Riverside dog park?", "which ordinances mention Canal Road?" — and get back the most relevant documents first, with a highlighted snippet showing why each one matched.

The `city_documents` table holds all three document types in one place, with a `body` column of plain prose text and a few relational columns for filtering:

Column	Purpose
`doc_type`	`council_minutes`, `zoning_ordinance`, or `public_notice`
`department`	Which city department published it (City Council, Public Works, …)
`title`	Short document title
`body`	Full plain-text document body — the field full-text search runs over
`published_date`	When the document was published

The documents share recurring topics on purpose — the Riverside dog park, the Harbour District waterfront rezoning, the Canal Road bike lane — so that searches in the exercises return more than one hit, and ranking has something real to sort between. No extensions are required for this chapter: `tsvector`, `tsquery`, and every function used below are core PostgreSQL.

Exercise Goals

By the end of this chapter you will be able to:

- Convert text to a `tsvector` and read the lexeme:position notation it produces.
- Explain exactly what stopword removal and stemming do to a document, using `ts_debug` to see PostgreSQL's tokenizer classify each word.
- Maintain a `tsvector` column automatically with a trigger, and index it with GIN for sub-millisecond searches.

- Write boolean queries with `to_tsquery` and understand why `plainto_tsquery` is the safer choice for a plain search box.
- Rank results with `ts_rank` and `ts_rank_cd`, and generate highlighted snippets with `ts_headline`.
- Build a custom text search configuration that treats domain-specific words as stopwords, and see it change real search results.

Installation

This chapter needs nothing beyond what Chapter 1 already set up: PostgreSQL 16 and a Python 3.12 virtual environment with `psycopg`. If you skipped Chapter 1, see its Installation section. Full-text search is core PostgreSQL — there is no extension to enable.

Loading the Data

Run the seed script

From the `book/` directory, with the virtual environment active:

```
python data/ch04_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema …
Inserting 30 documents …
Done — 30 rows in city_documents.
```

The seed script is self-contained — it does not depend on any earlier chapter's data. Note that it deliberately does **not** create a `tsvector` column or index; you build both by hand in Exercise 3, which is the point of the chapter.

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d city_documents
```

```
                                Table "public.city_documents"
     Column      |  Type   | Collation | Nullable
|                  Default
----------------+---------+-----------+----------
+------------------------------------------
 id              | integer |           | not null |
nextval('city_documents_id_seq'::regclass)
 doc_type        | text    |           | not null |
 department      | text    |           | not null |
 title           | text    |           | not null |
 body            | text    |           | not null |
 published_date | date    |           | not null |
Indexes:
    "city_documents_pkey" PRIMARY KEY, btree (id)
    "idx_city_documents_doc_type" btree (doc_type)
    "idx_city_documents_published_date" btree (published_date)
Check constraints:
    "city_documents_doc_type_check" CHECK (doc_type = ANY
(ARRAY['council_minutes'::text, 'zoning_ordinance'::text,
'public_notice'::text]))
```

Check 2 — counts by document type:

```sql
SELECT doc_type, COUNT(*) AS documents
FROM   city_documents
GROUP  BY doc_type
ORDER  BY doc_type;
```

```
     doc_type     | documents
------------------+-----------
 council_minutes  |        10
 public_notice    |        10
 zoning_ordinance |        10
(3 rows)
```

Check 3 — counts by department:

```sql
SELECT department, COUNT(*) AS documents
FROM   city_documents
GROUP  BY department
ORDER  BY department;
```

```
     department     | documents
--------------------+-----------
 City Council       |         9
 Finance            |         1
 Parks & Recreation |         3
 Planning & Zoning  |        12
 Public Works       |         5
(5 rows)
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — Converting Text to `tsvector`

1.1 — A basic conversion

`to_tsvector` is the function that turns plain text into PostgreSQL's searchable representation. Try it on one document's body:

```
SELECT title FROM city_documents WHERE id = 1;
```

```
              title
-----------------------------------
 Council Minutes — Harbour District Waterfront Renovation Budget
```

```
SELECT to_tsvector('english', body) FROM city_documents WHERE id =
1;
```

```
'along':38 'approv':67 'begin':79 'boardwalk':34 'bond':53
'budget':10 'citi':3
'construct':76 'conven':5 'council':4,41,61 'cover':28 'debat':56
'director':18
'discuss':43 'district':14 'expect':77 'fall':82 'first':69
'fund':44,74
'grant':50 'harbour':13,40,72 'includ':46 'infrastructur':49
'light':36
'measur':54 'member':42 'new':32 'one':65 'pedestrian':33 'phase':
26,70
'plan':27 'portsmith':2 'present':22 'propos':9 'public':20
'renov':16,73
'repair':30 'review':7 'seawal':29 'six':63 'sourc':45 'state':48
'three':25 'three-phas':24 'timelin':59 'upgrad':37 'vote':62
'waterfront':15 'work':21
```

1.2 — Reading the format

Each entry is `'lexeme':position,position,…`. A **lexeme** is a normalized word form, not the literal text — notice `'vote':62` even though the source text says "voted", `'approv':67` for "approve", and `'renov':16,73` for both "renovation" (position 16) and "renovation" again later (position 73, from "harbour **renovation** funding"). PostgreSQL stems words to their root so that a search for "vote" also matches documents containing "voted", "votes", or "voting" — you don't have to enumerate every inflection yourself.

The **positions** are the word's ordinal location in the original text (1st word, 2nd word, …). They exist for two reasons: `ts_rank` uses them to compute how spread out or clustered a document's matches are, and phrase search (`<->`, used in Exercise 4) uses them to require words to appear adjacent to each other, not just anywhere in the document.

1.3 — Case and punctuation are already handled

Note that "Council" (capitalized, position 4) and "council" (lowercase, positions 41 and 61) both collapsed into the single lexeme `'council'` with three positions — `to_tsvector` lowercases everything and strips punctuation as part of tokenization, before stemming ever runs.

Exercise 2 — What Stopword Removal Actually Throws Away

2.1 — Compare `simple` and `english` configurations

PostgreSQL ships several **text search configurations** — named bundles of tokenizer + dictionary rules. `simple` lowercases and tokenizes but does *no* stemming and drops *no* stopwords; `english` does both. Run the same body through each and count the resulting lexemes:

```
        SELECT array_length(regexp_split_to_array(body, '\s+'), 1) AS
raw_words
        FROM   city_documents WHERE id = 1;

 raw_words
-----------
        80

        SELECT array_length(tsvector_to_array(to_tsvector('simple', body)),
1) AS lexemes_simple,
               array_length(tsvector_to_array(to_tsvector('english', body)),
1) AS lexemes_english
        FROM   city_documents WHERE id = 1;

 lexemes_simple | lexemes_english
----------------+-----------------
              59 |              49
```

80 raw whitespace-separated words collapse to 59 distinct lexemes under `simple` (repeated words like "the" and "council" count once each, but nothing is removed or stemmed) and to 49 under `english` — ten fewer, because `english` additionally discards stopwords like "the", "a",

"of", "to", "in", and "and" entirely. They carry no search-relevant meaning on their own, and indexing them would only bloat the index with entries that match nearly every document.

2.2 — Watch the tokenizer classify each word with `ts_debug`

`ts_debug` is the diagnostic function for seeing exactly what a configuration does to a piece of text, word by word — useful whenever a search isn't matching what you expect and you need to know why. Run it on a short sentence:

```sql
SELECT alias, token, dictionaries, lexemes
FROM   ts_debug('english', 'The council voted to approve the budget for the harbour renovation.')
WHERE  alias <> 'blank';
```

```
   alias   |   token    |  dictionaries  |  lexemes
-----------+------------+----------------+-----------
 asciiword | The        | {english_stem} | {}
 asciiword | council    | {english_stem} | {council}
 asciiword | voted      | {english_stem} | {vote}
 asciiword | to         | {english_stem} | {}
 asciiword | approve    | {english_stem} | {approv}
 asciiword | the        | {english_stem} | {}
 asciiword | budget     | {english_stem} | {budget}
 asciiword | for        | {english_stem} | {}
 asciiword | the        | {english_stem} | {}
 asciiword | harbour    | {english_stem} | {harbour}
 asciiword | renovation | {english_stem} | {renov}
```

Every token gets classified (`asciiword` here — a plain word) and routed to the `english_stem` dictionary. Content words come back with a stemmed lexeme; stopwords ("The", "to", "the", "for", "the") come back with an **empty** `lexemes` array — recognized, looked up, and explicitly discarded. That empty array is the entire mechanism: a word is a stopword precisely when its dictionary entry maps it to nothing.

2.3 — When you'd reach for `simple` instead

`english` is the right default for prose. `simple` is useful for columns where stemming would be actively wrong — product SKUs, tag lists, or anything where "waterfront" and "waterfronts" should *not* be treated as the same token. `city_documents.body` is prose, so the rest of this chapter uses `english`.

Exercise 3 — A Maintained Column and a GIN Index

Computing `to_tsvector(body)` at query time works, but it means recomputing the same tokenization on every single search, over every row, every time. The standard pattern is to store the vector in its own column, keep it current as rows change, and index it.

3.1 — Add the column and backfill it

```
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;

UPDATE city_documents
SET    search_vector = to_tsvector('english', title || ' ' ||
body);
```

Indexing `title` alongside `body` means a search for a word that only appears in the title (not repeated in the body text) still matches.

3.2 — Keep it current with a trigger

A column populated once goes stale the moment anyone edits `title` or `body`. A `BEFORE` trigger recomputes it on every write:

```
CREATE FUNCTION city_documents_search_vector_update() RETURNS
trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' '
|| NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();
```

`OF title, body` scopes the trigger so it only fires when one of the two source columns actually changes — an `UPDATE` that only touches `published_date` doesn't pay for a needless re-tokenization.

> **Note — the manual way vs. the modern way:** This trigger pattern is how every PostgreSQL version has supported derived `tsvector` columns, and it's still exactly right when the derived value depends on more than one column, as it does here (`title` **and** `body`). PostgreSQL 12 added **generated columns** (`GENERATED ALWAYS AS (...) STORED`), which handle the common case — a `tsvector` derived from a single column — with less boilerplate and no trigger function to maintain by hand. Chapter 16 revisits

> this exact table and replaces this trigger with a generated column once `body` alone is the source. Both approaches produce an identical `tsvector`; the trigger is simply the general-purpose tool that works whenever the derivation is more than one column deep.

3.3 — Index it with GIN

```
CREATE INDEX idx_city_documents_search_vector
    ON city_documents USING GIN (search_vector);
```

```
\d city_documents
```

```
                                Table "public.city_documents"
     Column      |   Type    | Collation | Nullable
|                  Default
----------------+----------+----------+----------
+---------------------------------------------
 id              | integer  |          | not null |
nextval('city_documents_id_seq'::regclass)
 doc_type        | text     |          | not null |
 department      | text     |          | not null |
 title           | text     |          | not null |
 body            | text     |          | not null |
 published_date  | date     |          | not null |
 search_vector   | tsvector |          |          |
Indexes:
    "city_documents_pkey" PRIMARY KEY, btree (id)
    "idx_city_documents_doc_type" btree (doc_type)
    "idx_city_documents_published_date" btree (published_date)
    "idx_city_documents_search_vector" gin (search_vector)
Check constraints:
    "city_documents_doc_type_check" CHECK (doc_type = ANY
(ARRAY['council_minutes'::text, 'zoning_ordinance'::text,
'public_notice'::text]))
Triggers:
    trg_city_documents_search_vector BEFORE INSERT OR UPDATE OF
title, body ON city_documents FOR EACH ROW EXECUTE FUNCTION
city_documents_search_vector_update()
```

3.4 — Confirm the index is used

The `@@` operator tests whether a `tsvector` matches a `tsquery`:

```
EXPLAIN (ANALYZE, BUFFERS)
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ to_tsquery('english', 'harbour &
waterfront');
```

```
                                              QUERY PLAN
------------------------------------------------------------------
 Seq Scan on city_documents  (cost=0.00..8.38 rows=1 width=36)
(actual time=0.025..0.047 rows=5 loops=1)
   Filter: (search_vector @@ '''harbour'' &
''waterfront'''::tsquery)
   Rows Removed by Filter: 25
   Buffers: shared hit=8
```

On 30 rows the planner reasonably decides a sequential scan is cheaper than an index lookup — same behaviour you saw with the JSONB GIN index in Chapter 1. Force the index to confirm it works, exactly as in that chapter:

```sql
        SET enable_seqscan = off;

        EXPLAIN (ANALYZE, BUFFERS)
        SELECT id, title
        FROM   city_documents
        WHERE  search_vector @@ to_tsquery('english', 'harbour &
waterfront');

        SET enable_seqscan = on;   -- always restore this
```

```
Query Plan
------------------------------------------------------------------
 Bitmap Heap Scan on city_documents  (cost=12.98..16.99 rows=1
width=36) (actual time=0.024..0.033 rows=5 loops=1)
   Recheck Cond: (search_vector @@ '''harbour'' &
''waterfront'''::tsquery)
   Heap Blocks: exact=5
   Buffers: shared hit=10
   ->  Bitmap Index Scan on idx_city_documents_search_vector
(cost=0.00..12.98 rows=1 width=0) (actual time=0.014..0.014 rows=5
loops=1)
         Index Cond: (search_vector @@ '''harbour'' &
''waterfront'''::tsquery)
         Buffers: shared hit=5
```

`Bitmap Index Scan on idx_city_documents_search_vector` confirms the GIN index is doing the work. This is what makes full-text search viable at scale: instead of tokenizing every row's text on every query, PostgreSQL looks up each query lexeme directly in the index.

Exercise 4 — `to_tsquery` vs. `plainto_tsquery`

4.1 — `to_tsquery`: a boolean query language

`to_tsquery` expects an expression using explicit operators: `&` (AND), `|` (OR), `!` (NOT), and `<->` (phrase — “immediately followed by”):

```
        SELECT id, title
        FROM   city_documents
        WHERE  search_vector @@ to_tsquery('english', 'harbour &
waterfront')
        ORDER  BY id;
```

```
 id |                                 title
----
+-----------------------------------------------------------------
  1 | Council Minutes — Harbour District Waterfront Renovation
Budget
  8 | Council Minutes — Harbour District Food Truck Permits
 12 | Zoning Ordinance — Harbour District Waterfront Height
Variance
 22 | Public Notice — Public Hearing on Harbour District
Waterfront Rezoning
 30 | Public Notice — Annual Fireworks Display and Street Closures
(5 rows)
```

`!` excludes a term. This finds documents about flooding that are *not* about the flood-mitigation infrastructure program specifically:

```
        SELECT id, title
        FROM   city_documents
        WHERE  search_vector @@ to_tsquery('english', 'flood & !
mitigation')
        ORDER  BY id;
```

```
 id |                         title
----+-----------------------------------------------------
  5 | Council Minutes — Riverside Road Resurfacing Program
(1 row)
```

That single hit mentions “flooding” (which stems to `flood`) while discussing drainage damage, but never says “mitigation” — exactly what the query asked for.

4.2 — `to_tsquery` has no forgiveness for plain text

Feed it a raw phrase with no operator between the words, and it errors instead of guessing what you meant:

```
        SELECT id, title
        FROM   city_documents
        WHERE  search_vector @@ to_tsquery('english', 'flood mitigation');
```

```
ERROR:  syntax error in tsquery: "flood mitigation"
```

This is `to_tsquery`'s defining trade-off: it is a precise query language for code that constructs queries deliberately, and it is the wrong function to hand raw user input from a search box — a user typing "flood mitigation" with no operators will get an error page instead of results.

4.3 — `plainto_tsquery`: safe for a search box

`plainto_tsquery` takes plain text, tokenizes and stems it exactly like `to_tsvector` does, drops stopwords, and **ANDs** everything that survives. Feed it the same two words that made `to_tsquery` error out in 4.2, but typed the way an actual user would type them — with a stray "the" in front:

```
SELECT plainto_tsquery('english', 'the flood mitigation');
```

```
  plainto_tsquery
--------------------
 'flood' & 'mitig'
```

"the" never makes it into the query — `plainto_tsquery` drops it as a stopword, exactly the way `to_tsvector` would drop it from a document, and ANDs together whatever content words are left. Run it against the table:

```
SELECT id, title
FROM   city_documents
WHERE  search_vector @@ plainto_tsquery('english', 'the flood mitigation')
ORDER  BY id;
```

```
 id |                                     title
----
+------------------------------------------------------------------
 10 | Council Minutes — Special Session on Flood Mitigation
Infrastructure
 25 | Public Notice — City Council Special Session on Flood
Mitigation
(2 rows)
```

No error this time, and no operators to get wrong — `plainto_tsquery` took a sentence fragment with a stopword in it and produced exactly the query 4.2 had to write by hand. The trade-off runs the other way from `to_tsquery`: you get safety and stopword handling for free, but you lose the ability to express OR, NOT, or phrase search — `plainto_tsquery` always ANDs every surviving word together.

4.4 — | for OR searches

```
SELECT to_tsquery('english', 'dog | bike');
```

```
   to_tsquery
----------------
 'dog' | 'bike'

        SELECT id, title
        FROM   city_documents
        WHERE  search_vector @@ to_tsquery('english', 'dog | bike')
        ORDER  BY id;

 id |                               title
----
+------------------------------------------------------------------
  5 | Council Minutes — Riverside Road Resurfacing Program
  6 | Council Minutes — Riverside Dog Park Funding
  9 | Council Minutes — Canal Road Bike Lane Expansion
 15 | Zoning Ordinance — Reduced Parking Minimums Near Transit
Corridors
 24 | Public Notice — Riverside Dog Park Ribbon-Cutting Event
 28 | Public Notice — Canal Road Bike Lane Construction Schedule
 30 | Public Notice — Annual Fireworks Display and Street Closures
(7 rows)
```

Document 5 (road resurfacing) and 15 (parking minimums) show up because each mentions “bike” in passing — the resurfacing minutes note the work being coordinated with “the planned Canal Road bike lane construction,” and the parking ordinance references “the city’s ongoing bike lane expansion.” `OR` genuinely means *either*, which is exactly why it’s useful and why it returns more, looser matches than `AND`.

Exercise 5 — Ranking with `ts_rank` and `ts_rank_cd`, and `ts_headline` Snippets

`@@` tells you whether a document matches — it says nothing about *how well*. A user searching a 30-document archive can eyeball every hit, but a user searching 30,000 documents needs the best matches first.

5.1 — `ts_rank`: weight by term frequency

```
        SELECT id, title, round(ts_rank(search_vector, query)::numeric, 4)
AS rank
        FROM   city_documents, to_tsquery('english', 'harbour &
waterfront') AS query
        WHERE  search_vector @@ query
        ORDER  BY rank DESC;

 id |
title                                      |  rank
----
```

```
+---------------------------------------------------------------------
+--------
 12 | Zoning Ordinance — Harbour District Waterfront Height
Variance         | 0.1986
  1 | Council Minutes — Harbour District Waterfront Renovation
Budget        | 0.1959
 22 | Public Notice — Public Hearing on Harbour District
Waterfront Rezoning | 0.1895
  8 | Council Minutes — Harbour District Food Truck
Permits                  | 0.0999
 30 | Public Notice — Annual Fireworks Display and Street
Closures           | 0.0992
(5 rows)
```

`ts_rank` scores primarily on how often the query terms appear relative to the document's overall length — it does not care where in the document they appear or how close together they are. Documents 12, 1, and 22 rank highest because "harbour" and "waterfront" are central, repeated themes in short documents; 8 and 30 rank lower because each mentions the harbour only in passing within a longer document about something else.

5.2 — `ts_rank_cd`: weight by proximity too ("cover density")

```sql
        SELECT id, title,
               round(ts_rank(search_vector, query)::numeric, 4) AS rank,
               round(ts_rank_cd(search_vector, query)::numeric, 4) AS
rank_cd
        FROM   city_documents, to_tsquery('english', 'harbour &
waterfront') AS query
        WHERE  search_vector @@ query
        ORDER  BY rank_cd DESC;
```

```
 id |
title                                  |  rank  | rank_cd
----
+---------------------------------------------------------------------
+--------+---------
  1 | Council Minutes — Harbour District Waterfront Renovation
Budget        | 0.1959 |  0.1107
 22 | Public Notice — Public Hearing on Harbour District
Waterfront Rezoning | 0.1895 |  0.1053
 12 | Zoning Ordinance — Harbour District Waterfront Height
Variance         | 0.1986 |  0.0700
 30 | Public Notice — Annual Fireworks Display and Street
Closures           | 0.0992 |  0.0545
  8 | Council Minutes — Harbour District Food Truck
Permits                  | 0.0999 |  0.0542
(5 rows)
```

The order changes: document 12 had the *highest* `ts_rank` but drops to third under `ts_rank_cd`. `ts_rank_cd` additionally rewards matched terms that cluster close together in the text ("cover density") — in documents 1 and 22, "harbour" and "waterfront" appear right next to each other ("Harbour District **Waterfront** Renovation…"); in document 12 they occur further apart. Neither function is universally "more correct" — `ts_rank` is the standard choice; `ts_rank_cd` is worth trying when word proximity itself signals relevance, as it often does for short phrase-like queries.

5.3 — Highlighted snippets with `ts_headline`

A ranked list of titles is useful; showing *why* a document matched, with the query terms highlighted in context, is what users actually expect from a search results page:

```
        SELECT ts_headline('english', body, to_tsquery('english',
'harbour & waterfront'),
                             'StartSel=**, StopSel=**, MaxWords=25,
MinWords=10')
        FROM   city_documents
        WHERE  id = 1;
```

```
 **Harbour** District **waterfront** renovation. The Director of
Public Works presented
```

`ts_headline` re-scans the original text (not the `tsvector`) looking for the query terms, extracts a window around the best-matching fragment bounded by `MinWords`/`MaxWords`, and wraps each match in `StartSel`/`StopSel` markers — `**…**` here, but in a web application you'd use `<mark>…</mark>` or similar. This is genuinely expensive compared to a `tsvector` lookup (it re-tokenizes the source text on every call), so use it only on the page of results you're actually displaying, never inside a `WHERE` clause.

Exercise 6 — A Custom Text Search Configuration for Domain Stopwords

6.1 — The problem: some "content" words carry no information here

This is a city document archive. The words "Portsmith", "city", and "council" appear constantly — they are structurally present in almost every record, the way "the" is present in almost every English sentence. Standard `english` stopword removal has no way to know that, because it is tuned for general English, not this specific corpus:

```
        SELECT count(*) FROM city_documents
        WHERE  search_vector @@ to_tsquery('english', 'city & council');

 count
-------
     8
```

Eight of thirty documents — over a quarter of the entire archive — "match" a query on words that describe almost nothing about what makes any one of them relevant. Left alone, these words dilute ranking scores and inflate plain-text queries with noise terms the searcher didn't mean to require.

6.2 — Build a stopword file that extends the standard list

A text search configuration's stopword list is a plain text file, one word per line, that PostgreSQL reads from its `tsearch_data` directory. Start from the existing `english.stop` file so you keep every standard English stopword, then append the domain-specific ones:

```
        PG_SHAREDIR=$(pg_config --sharedir)
        sudo bash -c "cat '$PG_SHAREDIR/tsearch_data/english.stop' >
'$PG_SHAREDIR/tsearch_data/portsmith_english.stop'"
        sudo bash -c "printf 'portsmith\ncity\ncouncil\n' >>
'$PG_SHAREDIR/tsearch_data/portsmith_english.stop'"
```

> Placing files here requires root, since `tsearch_data` lives under PostgreSQL's shared install directory rather than anything database-owned. If your platform's PostgreSQL package stores it elsewhere, `pg_config --sharedir` will always point at the right location.

6.3 — Wire the file into a dictionary and a configuration

A stopword file alone does nothing — it has to be attached to a **text search dictionary**, and that dictionary mapped into a **text search configuration** that queries can actually reference:

```
        CREATE TEXT SEARCH DICTIONARY portsmith_stem (
            TEMPLATE = snowball,
            LANGUAGE = english,
            STOPWORDS = portsmith_english
        );

        CREATE TEXT SEARCH CONFIGURATION public.portsmith_english (COPY =
pg_catalog.english);

        ALTER TEXT SEARCH CONFIGURATION portsmith_english
            ALTER MAPPING FOR asciiword, asciihword, hword_asciipart,
```

```
word, hword, hword_part
            WITH portsmith_stem;
```

`TEMPLATE = snowball` reuses PostgreSQL's standard English stemming algorithm — only the stopword list changes, not the stemming rules. `COPY = pg_catalog.english` starts the new configuration as an exact clone of `english` (same tokenizer, same handling of numbers, emails, URLs, …), and the `ALTER MAPPING` line is the one substitution: word tokens now route through `portsmith_stem` instead of the stock `english_stem` dictionary.

6.4 — Confirm the noise words are gone

```
        SELECT to_tsvector('portsmith_english', body) FROM city_documents
WHERE id = 25;
```

```
'affect':37 'area':27 'attend':44 'basin':59 'comment':47
'conven':11 'develop':61
'discuss':16 'drain':56 'encourag':42 'engin':53 'flood':17,40,68
'given':4
'herebi':3 'infrastructur':19 'last':65 'mitig':18 'northgat':24
'notic':1
'open':31 'present':50 'propos':60 'provid':46 'public':34
'resid':36
'respons':63 'retain':25 'retent':58 'riversid':21 'session':14,29
'special':13
'spring':39 'staff':48 'storm':55 'wall':26 'year':66
```

Compare to `to_tsvector('english', body)` on the same row, which still carries `'citi':8,52`, `'council':9`, and `'portsmith':7` — three entries present under `english` and absent under `portsmith_english`. Everything else is untouched, because only the stopword list changed.

6.5 — Watch a noise-only query collapse to nothing

```
        SELECT to_tsquery('portsmith_english', 'city & council');
```

```
NOTICE:  text-search query contains only stop words or doesn't
contain lexemes, ignored
 to_tsquery
------------

(1 row)
```

Under `portsmith_english`, "city" and "council" are *both* stopwords, so a query built from nothing else has nothing left to search for — PostgreSQL says so explicitly instead of silently matching everything (or nothing) for an unclear reason.

6.6 — See it fix a real plain-text search

This is the exercise's payoff: a resident searching for "Portsmith dog park council" — a completely natural way to phrase it — under `english`:

```
        SELECT plainto_tsquery('english', 'Portsmith dog park council');

             plainto_tsquery
-------------------------------------------
 'portsmith' & 'dog' & 'park' & 'council'

        SELECT id, title FROM city_documents
        WHERE  search_vector @@ plainto_tsquery('english', 'Portsmith dog
park council')
        ORDER  BY id;

 id |                          title
----+-------------------------------------------------------------
 24 | Public Notice – Riverside Dog Park Ribbon-Cutting Event
(1 row)
```

Only one hit. The council minutes that actually approved funding for the same dog park (document 6) never happens to use the literal word "Portsmith," so requiring it as an AND term silently excludes the single most relevant record. Under `portsmith_english`:

```
        SELECT plainto_tsquery('portsmith_english', 'Portsmith dog park
council');

 plainto_tsquery
-----------------
 'dog' & 'park'

        SELECT id, title FROM city_documents
        WHERE  search_vector @@ plainto_tsquery('portsmith_english',
'Portsmith dog park council')
        ORDER  BY id;

 id |                          title
----+-------------------------------------------------------------
  6 | Council Minutes – Riverside Dog Park Funding
 24 | Public Notice – Riverside Dog Park Ribbon-Cutting Event
(2 rows)
```

Stripped down to the two words that actually distinguish this search — `dog` and `park` — both relevant documents come back. This is the concrete argument for a custom configuration: it is not a cosmetic tweak, it changes which documents a real user actually finds.

> **Note:** the `search_vector` column built in Exercise 3 was populated with the stock `english` configuration, so it still contains `'citi'`, `'council'`, and `'portsmith'` lexemes. The comparisons

> above work because `@@` only needs the *query's* surviving lexemes to be a subset of the document's — extra lexemes in the document that the query no longer asks for are simply irrelevant. To fully adopt `portsmith_english` as the table's search configuration going forward, you would update the trigger from Exercise 3 to call `to_tsvector('portsmith_english', …)` and rebuild `search_vector` for existing rows.

Summary — What You Should Now Know

Tool	What it does
`to_tsvector(config, text)`	Tokenizes, lowercases, stems, and removes stopwords, producing a searchable `lexeme:position` vector
`to_tsquery(config, expr)`	Parses a boolean query language (`&`, `|`, `!`, `<->`); errors on plain text with no operators
`plainto_tsquery(config, text)`	Tokenizes plain text the same way as `to_tsvector`, then ANDs every surviving lexeme — safe for a search box
`@@`	Tests whether a `tsvector` matches a `tsquery`
`ts_debug(config, text)`	Shows exactly how each token was classified and which lexeme (if any) it produced — the tool for "why didn't this match?"
GIN index on a `tsvector` column	Turns `@@` from a sequential scan into an index lookup
`ts_rank` / `ts_rank_cd`	Score match quality by term frequency, or by term frequency **and** proximity ("cover density")
`ts_headline(config, text, query, options)`	Re-scans the source text and returns a highlighted snippet — expensive; use only on displayed results
`CREATE TEXT SEARCH DICTIONARY ... (STOPWORDS = ...) + CREATE TEXT SEARCH CONFIGURATION`	Build a custom configuration with domain-specific stopwords

The key design insight from this chapter is that full-text search is not one function call — it's a pipeline (tokenize → stem → filter stopwords) that you can inspect at every stage with `ts_debug`, store the output of with a maintained column and a GIN index, query against with two very different trade-offs (`to_tsquery` for precision, `plainto_tsquery` for safety), and tune for your specific corpus by swapping the stopword list — exactly the kind of tuning a generic search engine bolted on top of your database can't do without shipping your schema knowledge to it.

The `city_documents` table you built here is reused directly in later chapters: Chapter 6 (`pgvector`) adds embeddings to the same table for semantic search and builds a hybrid search combining `ts_rank` with cosine distance, and Chapter 16 (Generated Columns) replaces the hand-written trigger from Exercise 3 with a `GENERATED ALWAYS AS (...) STORED` column.

Going further: `websearch_to_tsquery` *(PostgreSQL 11+) is worth knowing about even though this chapter didn't use it — it accepts the same quoted-phrase and* `-exclude` *syntax as a typical web search box (e.g.* `"flood mitigation" -infrastructure`*) while remaining as forgiving as* `plainto_tsquery`*, making it the best default for a real user-facing search field. For very large document sets, look into PostgreSQL's built-in support for weighted vectors (*`setweight()`*, ranking title matches above body matches) and consider whether a dedicated search engine becomes worthwhile once ranking quality and query latency requirements outgrow what a GIN index over a single table can deliver — the honest answer, for most applications, is later than you'd expect.*

Chapter 5 — Fuzzy Matching: `pg_trgm`

> *"Every registry that outlives a year of real data entry ends up with the same person in it twice, spelled two different ways."*

Background

Full-text search, from the last chapter, is built for a specific kind of imprecision: the same *word*, inflected differently ("vote", "voted", "voting"). It does nothing for a different, equally common kind of imprecision — the same word, **spelled** differently. "Portsmith" typed as "Portsmith" versus "Portsmyth". "McAllister" versus "MacAllister". A resident registry filled in by hand, twice, by two different clerks, on two different days. Stemming cannot fix a typo, because a typo isn't a grammatical variant of the correct word — it's a different string that a human reader recognizes as "close enough" and a computer, by default, does not.

`pg_trgm` is PostgreSQL's answer: it breaks strings into overlapping three-character sequences (**trigrams**) and measures how many trigrams two strings share. Two spellings of the same name share most of their trigrams even when several letters differ; two unrelated strings usually share almost none. That single idea — measured, thresholded, and indexed — is enough to power "did you mean?" search boxes, deduplicate messy records, and accelerate substring `LIKE` queries that would otherwise force a sequential scan.

This chapter builds a small deduplication and search tool around that idea: what a trigram actually is, how `similarity()` scores a pair of strings, how to turn that into an index instead of an $O(n^2)$ scan, and — the harder, more honest part — where trigram matching's judgment calls are, because no similarity threshold gets every case right.

The Scenario

Portsmith's resident registry has been filled in over years by different staff at different counters, and it shows: the same person appears twice under two spellings of their name often enough that nobody trusts a raw `COUNT(*)` on it anymore. Separately, the city's 311 line fields phone searches for local businesses, and callers rarely spell a business name correctly on the first try.

Two tables model this:

Table	Purpose
`residents`	Synthetic residents, including intentional near-duplicate entries — the same person, typed twice, with a typo, transposition, or variant spelling the second time
`business_names`	A flat `(business_id, name)` lookup extending the `businesses` table from Chapter 1, used for "did you mean?" search

`residents` includes a `true_duplicate_of` column that a real registry would **not** have — you would not know in advance which rows are duplicates; that is the entire problem fuzzy matching exists to solve. It's here purely so the exercises can check whether your queries found the right answers. The dataset also includes two pairs that are deliberately *not* marked as duplicates, on purpose — more on those in Exercise 2.

Exercise Goals

By the end of this chapter you will be able to:

- Explain what a trigram is and compute `similarity()` between two strings by hand for a short example.
- Use the `%` operator to find near-matches at a configurable similarity threshold, and explain why no single threshold is perfectly correct.
- Use `word_similarity()` to find a short query as a strong partial match inside a longer string, and explain how it differs from `similarity()`.
- Create a GIN trigram index and confirm it accelerates both `%` and `LIKE '%term%'` queries that would otherwise force a sequential scan.

- Build a "did you mean?" query using a GiST trigram index and `ORDER BY ... <->`, and explain why GiST — not GIN — is the right index for that access pattern.
- Compare trigram matching against full-text search on the same kind of short, keyword-style query, and know which one to reach for.

Installation

`pg_trgm` ships as part of the standard `postgresql-16` package on Debian/Ubuntu — unlike PostGIS in Chapter 2, there is no separate package to install. Enable it in the database:

```sql
CREATE EXTENSION IF NOT EXISTS pg_trgm;
```

Loading the Data

Prerequisites

Chapter 1's seed script must have been run first — `business_names` is populated directly from the `businesses` table:

```
python data/ch01_seed.py
```

Run the Chapter 5 seed

```
python data/ch05_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Creating schema …
Inserting 58 residents …
Populating business_names from businesses …
Done — 58 rows in residents, 48 rows in business_names.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — table structure:

```
\d residents
```

```
                                  Table "public.residents"
      Column       |  Type    | Collation | Nullable
|                Default
-------------------+---------+-----------+----------
+--------------------------------------
 id                | integer |           | not null |
nextval('residents_id_seq'::regclass)
 full_name         | text    |           | not null |
 neighbourhood     | text    |           | not null |
 true_duplicate_of | integer |           |          |
Indexes:
    "residents_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "residents_true_duplicate_of_fkey" FOREIGN KEY
(true_duplicate_of) REFERENCES residents(id)
Referenced by:
    TABLE "residents" CONSTRAINT
"residents_true_duplicate_of_fkey" FOREIGN KEY (true_duplicate_of)
REFERENCES residents(id)
```

Check 2 — every marked duplicate points at a real canonical row:

```sql
SELECT COUNT(*) AS duplicate_pairs
FROM   residents
WHERE  true_duplicate_of IS NOT NULL;
```

```
 duplicate_pairs
-----------------
              12
```

Check 3 — `business_names` mirrors `businesses` 1:1:

```sql
SELECT COUNT(*) FROM business_names;
```

```
 count
-------
    48
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — What a Trigram Is, and `similarity()`

1.1 — Break a string into trigrams

`show_trgm()` returns the actual set of trigrams PostgreSQL generates for a string:

```sql
SELECT show_trgm('Eleanor');
```

```
                show_trgm
-------------------------------------------
 {"  e"," el",ano,ean,ele,lea,nor,"or "}
```

Before splitting, PostgreSQL pads the string with two leading spaces and one trailing space — `"  eleanor "` — then takes every overlapping run of three characters: `"  e"`, `" el"`, `"ele"`, `"lea"`, `"ean"`, `"ano"`, `"nor"`, `"or "`. The padding matters: it means the first and last letters of a word each get their own distinguishing trigram (`"  e"` marks "starts with e"; `"or "` marks "ends with or"), so two strings that only differ at the very start or end still register as different rather than accidentally looking identical in the middle.

1.2 — `similarity()`: how much overlap, as a fraction

```sql
SELECT similarity('Eleanor Whitmore', 'Elenor Whitmore');
```

```
 similarity
------------
  0.7368421
```

`similarity()` compares the trigram sets of both strings and returns roughly the fraction that overlap — mostly shared trigrams (both names are 16-17 characters, differing by one dropped letter) means a high score close to 1. Two unrelated strings — "Eleanor Whitmore" and, say, "Ironside Auto" — share almost no trigrams and score close to 0. The scale is intuitive by construction: 1.0 is identical, 0.0 is nothing alike.

1.3 — All twelve duplicate pairs, scored

```sql
SELECT a.full_name AS canonical, b.full_name AS duplicate_entry,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS
sim
FROM   residents a
JOIN   residents b ON b.true_duplicate_of = a.id
ORDER  BY a.id;
```

```
      canonical         |    duplicate_entry     |  sim
------------------------+------------------------+-------
 Eleanor Whitmore       | Elenor Whitmore        | 0.737
 Jonathan Castellano    | Jonathon Castellano    | 0.739
 Priyanka Deshmukh      | Priyanka Deshmuk       | 0.842
 Bartholomew Okonkwo    | Bartholemew Okonkwo    | 0.739
 Marguerite Delacroix   | Marguerite Delacroiux  | 0.792
```

```
 Siobhan McAllister      | Siobhan MacAllister     | 0.773
 Theodore Vance          | Theodor Vance           | 0.813
 Anastasia Volkov        | Anastassia Volkov       | 0.842
 Desmond Okafor          | Desmund Okafor          | 0.667
 Genevieve Laurent       | Genevieve Lorent        | 0.667
 Mikhail Petrenko        | Mikail Petrenko         | 0.737
 Fitzgerald Osei         | Fitzgerld Osei          | 0.722
```

Every genuine duplicate in this dataset scores well above 0.6, regardless of whether the typo was a dropped letter, a swapped letter, or a spelling variant. That range is the working intuition Exercise 2 turns into an actual threshold.

Exercise 2 — The `%` Operator, and Why One Threshold Isn't Enough

2.1 — `%`: "similar enough," using a session-wide threshold

Computing `similarity()` for every pair by hand doesn't scale. The `%` operator wraps it into a boolean test against a configurable cutoff:

```
SHOW pg_trgm.similarity_threshold;
```

```
 pg_trgm.similarity_threshold
------------------------------
 0.3
```

`0.3` is the default. Self-join `residents` against itself to find every pair of *different* rows that clears it — this is the actual "find likely duplicates" query:

```
SELECT a.id, a.full_name, b.id, b.full_name,
       round(similarity(a.full_name, b.full_name)::numeric, 3) AS
sim
FROM   residents a
JOIN   residents b ON a.id < b.id
WHERE  a.full_name % b.full_name
ORDER  BY sim DESC, a.id;
```

```
 id |       full_name       | id |       full_name        |  sim
----+-----------------------+----+------------------------+-------
 35 | Priyanka Deshmukh     | 36 | Priyanka Deshmuk       | 0.842
 45 | Anastasia Volkov      | 46 | Anastassia Volkov      | 0.842
 43 | Theodore Vance        | 44 | Theodor Vance          | 0.813
 39 | Marguerite Delacroix  | 40 | Marguerite Delacroiux  | 0.792
 41 | Siobhan McAllister    | 42 | Siobhan MacAllister    | 0.773
 57 | Nadia Kowalski        | 58 | Nadia Kowalska         | 0.765
 33 | Jonathan Castellano   | 34 | Jonathon Castellano    | 0.739
 37 | Bartholomew Okonkwo   | 38 | Bartholemew Okonkwo    | 0.739
```

```
 31 | Eleanor Whitmore       | 32 | Elenor Whitmore         | 0.737
 51 | Mikhail Petrenko       | 52 | Mikail Petrenko         | 0.737
 53 | Fitzgerald Osei        | 54 | Fitzgerld Osei          | 0.722
 47 | Desmond Okafor         | 48 | Desmund Okafor          | 0.667
 49 | Genevieve Laurent      | 50 | Genevieve Lorent        | 0.667
 55 | Robert Ashworth        | 56 | Bobby Ashworth          | 0.409
(14 rows)
```

`a.id < b.id` keeps each pair once instead of twice (A-vs-B and B-vs-A are the same comparison). Notice: **fourteen** rows came back, not twelve. Check `true_duplicate_of` on every row involved and you'll find twelve of these pairs marked as genuine duplicates and two that are not — "Nadia Kowalski" / "Nadia Kowalska" and "Robert Ashworth" / "Bobby Ashworth". They're in this dataset on purpose, and — this is the point of the exercise — look at *where* they landed in the ranking.

2.2 — A false positive sitting in the middle of the true positives

"Nadia Kowalski" and "Nadia Kowalska" are two unrelated Riverside residents who happen to share a first name and a Polish surname that differs only in its masculine/feminine ending. Their similarity, `0.765`, isn't an edge case at the bottom of the list — it sits **between** "Siobhan McAllister"/"Siobhan MacAllister" (`0.773`) and "Bartholomew Okonkwo"/ "Bartholemew Okonkwo" (`0.739`), two genuine duplicates. `similarity()` is doing exactly what it's designed to do: these two strings really are that close. Whether that means "probably the same person" is a judgment call the function cannot make on its own — the central limitation of fuzzy matching. It finds **candidates**, not verified duplicates. A production dedup pipeline treats a `%` match as "flag for review" or "compare a second field too" (a shared phone number or address), never as "merge automatically."

2.3 — Prove no threshold fixes it

Try to set a threshold that excludes the Kowalski pair:

```sql
SET pg_trgm.similarity_threshold = 0.77;
```

Re-run the query from 2.1. "Nadia Kowalski"/"Nadia Kowalska" (`0.765`) is gone — but so are seven of the twelve genuine duplicates, everything scoring below `0.77`. There is no number you can put in that `SET` statement that keeps all twelve real duplicates and excludes the Kowalski pair, because a real duplicate ("Bartholomew Okonkwo"/"Bartholemew Okonkwo", `0.739`) scores *lower* than the false positive you're trying to exclude. Put the threshold back before continuing:

```sql
SET pg_trgm.similarity_threshold = 0.3;
```

2.4 — A different failure: the false negative you'd never even see

"Robert Ashworth" and "Bobby Ashworth" are, in this dataset's backstory, the same person — entered once with a nickname. Notice they only made the results list at all (`0.409`) because of the shared surname "Ashworth"; as given names alone, "Robert" and "Bobby" share nothing:

```
        SELECT similarity('Robert', 'Bobby');

 similarity
------------
          0
```

A resident with a *less* distinctive shared surname and a nicknamed given name would score near zero overall and never appear in a `%` results list at any threshold — not a borderline case to review, just silently absent. `similarity()` measures string closeness, not identity; it has no way to know "Bobby" is short for "Robert" unless something else tells it, such as a synonym table consulted alongside it.

2.5 — The honest takeaway

Between the Kowalski pair and the Ashworth pair, this dataset has one false positive that no threshold can cleanly exclude without also losing real duplicates, and one true duplicate that scores so low it would never surface as a candidate in the first place. Tune `pg_trgm.similarity_threshold` to trade recall against precision for your own data — but go in expecting a trade-off, not a number that gets both sides to zero.

Exercise 3 — `word_similarity()` for Partial Matches

`similarity()` compares two whole strings — it penalizes a short query against a long target just for being different lengths, which makes it a poor fit for "does this short input appear as a strong match somewhere inside this longer string?"

3.1 — Compare the two functions on the same query

```
        SELECT name, round(similarity('bake', name)::numeric, 3) AS
full_sim
        FROM   business_names
        ORDER  BY full_sim DESC, name
        LIMIT  3;
```

```
          name           | full_sim
-------------------------+----------
 River Bend Bakery       |    0.222
 Campus Bike & Sports    |    0.091
 Finch & Sons Barbers    |    0.091
```

```
        SELECT name, round(word_similarity('bake', name)::numeric, 3) AS
word_sim
        FROM   business_names
        ORDER  BY word_sim DESC, name
        LIMIT  3;
```

```
          name           | word_sim
-------------------------+----------
 River Bend Bakery       |    0.800
 Bay Street Electronics  |    0.400
 Finch & Sons Barbers    |    0.400
```

Same query, same top result, very different score: `0.222` versus `0.800`. `word_similarity()` finds the best-matching *substring extent* inside the target — effectively "if I could crop this longer string down to the part that best matches my query, how similar would that piece be?" — rather than scoring the query against the target's full length. "bake" against "River Bend **Bake**ry" scores high because "Bake" is a near-perfect fragment match, even though "bake" is a small fraction of the full business name.

3.2 — When to reach for which

Use `similarity()` when you're comparing two values that *should* represent the same whole thing — two spellings of one name, as in Exercises 1 and 2. Use `word_similarity()` when a short query is expected to be a fragment of a longer field — autocomplete-style search-as-you-type, or matching a partial business name a caller remembers.

Exercise 4 — GIN Trigram Indexes

4.1 — Without an index

```
        EXPLAIN (ANALYZE, BUFFERS)
        SELECT name FROM business_names WHERE name LIKE '%Bakery%';
```

```
                                                QUERY PLAN
-----------------------------------------------------------------
 Seq Scan on business_names  (cost=0.00..25.88 rows=1 width=32)
(actual time=0.008..0.009 rows=1 loops=1)
   Filter: (name ~~ '%Bakery%'::text)
```

```
   Rows Removed by Filter: 47
   Buffers: shared hit=1
```

A plain B-tree index cannot help here — `%Bakery%` has no fixed prefix, so there's nothing for a B-tree to seek to. This is exactly the case `pg_trgm` was built for: it indexes every trigram in every row, so a `LIKE` pattern (itself broken into trigrams) can be looked up directly.

4.2 — Create the index

```sql
CREATE INDEX idx_business_names_trgm
    ON business_names
    USING GIN (name gin_trgm_ops);
```

4.3 — Confirm it's used

As in earlier chapters, force the index on a table this small to see the mechanism:

```sql
SET enable_seqscan = off;

EXPLAIN (ANALYZE, BUFFERS)
SELECT name FROM business_names WHERE name LIKE '%Bakery%';

SET enable_seqscan = on;    -- always restore this
```

```
                                                            QUERY
PLAN
-------------------------------------------------------------------
 Bitmap Heap Scan on business_names  (cost=21.82..25.83 rows=1
width=32) (actual time=0.024..0.025 rows=1 loops=1)
   Recheck Cond: (name ~~ '%Bakery%'::text)
   Heap Blocks: exact=1
   Buffers: shared hit=10
   ->  Bitmap Index Scan on idx_business_names_trgm
(cost=0.00..21.82 rows=1 width=0) (actual time=0.016..0.017 rows=1
loops=1)
         Index Cond: (name ~~ '%Bakery%'::text)
         Buffers: shared hit=9
```

`Bitmap Index Scan on idx_business_names_trgm` — the same GIN index also accelerates `%`, `word_similarity()` `%>`, and both `LIKE` and `ILIKE` with leading wildcards, all from one index.

Exercise 5 — A "Did You Mean?" Query with GiST

5.1 — The pattern: order by trigram distance, take the top few

`pg_trgm` defines a distance operator, `<->` — the complement of similarity (smaller means more alike) — which makes "closest matches first" a plain `ORDER BY ... LIMIT`:

```
        SELECT name, round(similarity(name, 'Ironsyde Auto')::numeric, 3)
AS sim
        FROM   business_names
        ORDER  BY name <-> 'Ironsyde Auto', name
        LIMIT  5;

        name          |  sim
----------------------+-------
 Ironside Auto        | 0.647
 AutoFix Portsmith    | 0.143
 Harbour Inn          | 0.040
 The Art Depot        | 0.037
 Riverside Cinema     | 0.033
```

A caller who typed "Ironsyde Auto" (transposed letters) gets "Ironside Auto" back as the clear top match, well clear of the noise below it — this is the entire "did you mean?" feature.

5.2 — Why the GIN index from Exercise 4 doesn't help here

```
        EXPLAIN (ANALYZE)
        SELECT name FROM business_names
        ORDER BY name <-> 'Ironsyde Auto'
        LIMIT  5;

                                                      QUERY PLAN
------------------------------------------------------------------
 Limit  (cost=2.40..2.41 rows=5 width=36) (actual
time=0.182..0.183 rows=5 loops=1)
   ->  Sort  (cost=2.40..2.52 rows=48 width=36) (actual
time=0.181..0.182 rows=5 loops=1)
         Sort Key: ((name <-> 'Ironsyde Auto'::text))
         Sort Method: top-N heapsort  Memory: 25kB
         ->  Seq Scan on business_names  (cost=0.00..1.60 rows=48
width=36) (actual time=0.040..0.157 rows=48 loops=1)
```

Even with `idx_business_names_trgm` in place, this plan is a full scan plus a sort — GIN accelerates *lookups* ("which rows contain trigrams matching this pattern") but has no notion of a distance ordering. Nearest-neighbor queries need an index that natively understands "closest," which is what **GiST** provides:

```
        CREATE INDEX idx_business_names_trgm_gist
            ON business_names
            USING GIST (name gist_trgm_ops);
```

```sql
EXPLAIN (ANALYZE)
SELECT name FROM business_names
ORDER BY name <-> 'Ironsyde Auto'
LIMIT  5;
```

```
QUERY PLAN
-------------------------------------------------------------------
 Limit  (cost=0.14..1.07 rows=5 width=36) (actual
time=0.130..0.143 rows=5 loops=1)
   ->  Index Scan using idx_business_names_trgm_gist on
business_names  (cost=0.14..9.09 rows=48 width=36) (actual
time=0.129..0.141 rows=5 loops=1)
         Order By: (name <-> 'Ironsyde Auto'::text)
```

`Order By: (name <-> 'Ironsyde Auto'::text)` inside an `Index Scan` — the GiST index walks straight to the nearest rows instead of scoring and sorting every row in the table. **Rule of thumb: GIN for `%`/`LIKE` membership lookups, GiST for `ORDER BY ... <-> ... LIMIT n` nearest-match queries.** It's common to keep both, as this table now does, if an application needs both access patterns.

5.3 — One more example

```sql
        SELECT name, round(similarity(name, 'Portsmith Vetrinary
Clinic')::numeric, 3) AS sim
        FROM   business_names
        ORDER  BY name <-> 'Portsmith Vetrinary Clinic', name
        LIMIT  5;
```

```
            name                |  sim
--------------------------------+-------
 Portsmith Veterinary Clinic    | 0.833
 AutoFix Portsmith              | 0.286
 Portsmith Pharmacy             | 0.286
 Portsmith Tailors              | 0.286
 Portsmith Arms Hotel           | 0.263
```

Exercise 6 — Trigram Matching vs. Full-Text Search, Head to Head

Chapter 4 built full-text search over `city_documents`; this chapter built trigram matching over `business_names`. Both can answer "find rows matching this word" — they are good at it for different, almost opposite, reasons.

6.1 — A spelling variant: trigram wins

Portsmith's own documents consistently use the British spelling "harbour." A caller who types the American spelling "harbor" gets nothing from full-text search — stemming normalizes *inflection* ("harbors" → "harbor"), not *spelling*:

```
        SELECT to_tsvector('english', 'Harbour View Theater') @@
to_tsquery('english', 'harbor');

 ?column?
----------
 f
```

Trigram similarity doesn't care that they're "different words" — it only sees how many three-letter fragments overlap, and "harbor"/"harbour" share almost all of theirs:

```
        SELECT similarity('harbor', 'harbour');

 similarity
------------
        0.5
```

6.2 — A short, correctly-spelled keyword: full-text search wins

Run a short, exact keyword against `business_names` both ways:

```
        SELECT name, round(similarity(name, 'bay')::numeric, 3) AS sim
        FROM   business_names
        ORDER  BY sim DESC, name
        LIMIT  6;

          name            |  sim
--------------------------+-------
 Mango Bay Caribbean      | 0.200
 Bay Street Electronics   | 0.174
 River Bend Bakery        | 0.105
 Finch & Sons Barbers     | 0.095
 Bella Napoli             | 0.063
 Le Petit Bistro          | 0.053
```

A three-letter query like "bay" barely produces any trigrams of its own, so the scores are all low and close together — "River Bend Bakery" and "Finch & Sons Barbers" show up with real, if small, similarity despite having nothing to do with "bay." There's no clean gap between signal and noise. Full-text search, which matches on whole tokens rather than character fragments, has no such problem:

```
        SELECT name FROM business_names
        WHERE  to_tsvector('english', name) @@ plainto_tsquery('english',
'bay')
        ORDER  BY name;
```

```
          name
-----------------------
 Bay Street Electronics
 Mango Bay Caribbean
(2 rows)
```

Exactly the two relevant matches, no noise, no threshold to tune.

6.3 — The rule of thumb

Short queries made of correctly-spelled whole words belong to full-text search — it was built to match tokens precisely and cheaply via a GIN index over lexemes. Queries that might be misspelled, transposed, or OCR-damaged belong to trigram matching — it was built to tolerate exactly that kind of noise, at the cost of weaker signal on very short inputs. Many real search boxes run both: try full-text search first, and fall back to a trigram "did you mean?" only when it returns nothing.

Summary — What You Should Now Know

Tool	What it does
`show_trgm(text)`	Shows the padded, overlapping 3-character sequences a string breaks into
`similarity(a, b)`	Fraction of shared trigrams between two whole strings — best for "are these two values the same thing, spelled differently?"
`word_similarity(a, b)`	Best-matching substring extent of `a` inside `b` — best for "is `a` a fragment somewhere inside `b`?"
`a % b`	Boolean test: does `similarity(a, b)` clear `pg_trgm.similarity_threshold` (default `0.3`)?
`a <-> b`	Trigram distance (`1 - similarity`) — sort by this for nearest-match-first results
`GIN (col gin_trgm_ops)`	Accelerates `%` and `LIKE`/`ILIKE` membership-style lookups
`GIST (col gist_trgm_ops)`	Accelerates `ORDER BY col <-> query LIMIT n` nearest-neighbor lookups

Tool	What it does
`pg_trgm.similarity_threshold`	Session-level cutoff for `%`; tune it, but expect trade-offs, not a perfect split

The key design insight from this chapter is that fuzzy matching produces *candidates*, not verdicts. The `%` operator surfaced all twelve real duplicate pairs in the `residents` table — and two pairs of genuinely different, or differently-named, people right alongside them, at every threshold tried. That isn't a shortcoming to engineer away; it's the nature of measuring string similarity instead of identity. Build fuzzy matching into a review queue or a secondary confirmation step, not an automatic merge.

The `business_names` table you built here is reused in Chapter 10 (PostgREST), which exposes the "did you mean?" query from Exercise 5 as a public RPC endpoint.

Going further: `pg_trgm` *also provides* `%>` *and* `<->>` *variants tuned for* `word_similarity()` *rather than* `similarity()`*, useful for indexing substring-style "did you mean?" search the same way Exercise 5 indexed whole-string search. For very large tables, GiST trigram indexes are typically smaller and faster to build than GIN but slower for pure membership lookups — benchmark both on your actual data distribution rather than assuming one is strictly better. And if deduplication is a recurring, business-critical process rather than an occasional query, look at dedicated record-linkage tools (e.g. the* `dedupe` *Python library), which combine trigram-style string similarity with other fields — address, phone number, date of birth — and a trained classifier, instead of a single threshold on a single column.*

Chapter 6 — Vector Search: `pgvector` for Embeddings

> *"Full-text search finds documents that use your words. Vector search finds documents that share your meaning, whether or not they share a single word with you."*

Background

Chapter 4 matched documents by shared vocabulary. Chapter 5 matched strings by shared spelling. Neither technique can find a document about "the city's dog park" when a resident searches "somewhere my pet can run off-leash" — there isn't a misspelling to correct or a stemmed word in common; the two phrases are simply *about the same thing*, expressed in unrelated words. No amount of stemming, stopword tuning, or trigram threshold-adjusting closes that gap, because all three earlier techniques operate on the text itself. Closing it requires a representation of what the text *means*.

That representation is an **embedding**: a machine learning model reads a piece of text and outputs a vector — literally an array of a few hundred floating-point numbers — positioned in a high-dimensional space such that texts with similar meaning end up as nearby points and unrelated texts end up far apart. "Somewhere my pet can run off-leash" and "the city's dog park" land close together in that space even though they share zero words. This chapter never trains such a model — it uses a small, well-established, freely available one (`all-MiniLM-L6-v2`) to turn text into 384-number vectors, and PostgreSQL's `pgvector` extension to store, index, and search them.

The two hard problems `pgvector` solves are storage (a native `vector` type, instead of smuggling floats through a JSONB array or a comma-separated text column) and **search at scale**. Finding the closest vector to a query by brute force means comparing it against every single row — fine for the thirty documents in Chapter 4's table, hopeless for a million-row photo library. This chapter builds up to two different **approximate nearest neighbor (ANN)** index types, IVFFlat and HNSW, that trade a small amount of accuracy for a large amount of speed, and it's honest about exactly how much accuracy each one actually costs you at its defaults, because the answer is more surprising than most introductions let on.

The Scenario

Two tables carry this chapter, and they get their vectors two very different ways, on purpose.

`city_documents` — the council minutes, zoning ordinances, and public notices from Chapter 4 — gets a real `embedding` column computed by actually running `all-MiniLM-L6-v2` over each document's title and body. These are genuine embeddings: semantic search over them in Exercise 5 finds real conceptual matches, not a scripted demo.

`city_photos` is different, and says so in its own name: it's a synthetic stand-in for a photo library's image embeddings (the kind a real vision model like CLIP would produce for actual photographs), used purely to give Exercises 3 and 4 a table large enough — 5,000 rows across ten categories — for approximate indexing to be worth doing at all. Thirty rows, like `city_documents` has, will never be large enough to make an ANN index pay for itself; a sequential scan over thirty rows is already about as fast as computation gets. There is no real photo behind any row in `city_photos` — each one is a random point clustered around one of ten category "anchors" in 384-dimensional space, which reproduces the *shape* of a real embedding space (similar things cluster; different things don't) without needing a single actual image.

Table	Vectors	Used for
`city_documents`	Real — `all-MiniLM-L6-v2` embeddings of each document's title + body	Semantic search, hybrid search (Exercises 5-6)
`city_photos`	Synthetic — 5,000 rows, 10 clustered categories, no real images	ANN indexing at scale (Exercises 3-4)

Exercise Goals

By the end of this chapter you will be able to:

- Store embeddings in a native `vector` column and understand what the numbers in it actually represent.
- Compute exact nearest neighbors with `<->` (L2), `<#>` (negative inner product), and `<=>` (cosine distance), and know when the choice between them actually changes your results.
- Build an IVFFlat index, and correctly interpret its `lists` and `probes` parameters instead of trusting the defaults.
- Build an HNSW index, and make an informed build-time-vs-recall trade-off between it and IVFFlat instead of guessing.
- Run real semantic search: find documents by meaning even when they share no vocabulary with the query.
- Build hybrid search that blends a keyword score (`ts_rank`, from Chapter 4) with a semantic score into one ranking.

Installation

This chapter has two dependencies, and they're independent of each other — you need both, but you're installing two unrelated things for two unrelated reasons.

1 — `pgvector`

```
sudo apt install -y postgresql-16-pgvector
```

Enable the extension. **Unlike `pg_trgm` in Chapter 5, `pgvector` 0.6.x is not a "trusted" extension** — its control file doesn't set `trusted = true`, so a regular database-owning role cannot enable it with a plain `CREATE EXTENSION`. It has to be done once, by a superuser:

```
sudo -u postgres psql portsmith -c "CREATE EXTENSION vector;"
```

Confirm it's there:

```
SELECT extversion FROM pg_extension WHERE extname = 'vector';
```

```
 extversion
------------
 0.6.0
```

2 — `sentence-transformers`, CPU-only

The embedding model runs locally, in Python, via the `sentence-transformers` library — no API key, no network calls at query time. It depends on PyTorch, and installing PyTorch's default package pulls in a full NVIDIA CUDA toolkit (several gigabytes) even on a machine with no GPU, which is pure waste for embedding a few dozen short documents on CPU. Install the CPU-only build explicitly, then `sentence-transformers` on top of it:

```
python3.12 -m venv .venv    # if you haven't already, from earlier chapters
source .venv/bin/activate
pip install torch --index-url https://download.pytorch.org/whl/cpu
pip install sentence-transformers pgvector
```

> **Note:** this installs roughly 3GB into your virtual environment (PyTorch and its numerical dependencies are not small, even CPU-only) and the first time any script loads `all-MiniLM-L6-v2`, it downloads about 90MB of model weights from Hugging Face and caches them in `~/.cache/huggingface`. Every run after that is fully offline. If you see a `Warning: You are sending unauthenticated requests to the HF Hub` line the first time, that's expected and harmless — it's just Hugging Face reminding you that an account would raise your download rate limit.

Loading the Data

Prerequisites

Chapter 4's seed script must have been run first:

```
python data/ch04_seed.py
```

`ch04_seed.py` alone is **not** enough, though — this chapter's hybrid search (Exercise 6) needs `city_documents.search_vector`, and that column only gets created by hand-running Chapter 4's **Exercise 3** SQL; it is not part of `ch04_seed.py`'s schema. (If you re-run `ch04_seed.py` after already having done Exercise 3 — say, to reset the document data — it drops `city_documents` and rebuilds it *without* `search_vector`, silently undoing that exercise.) `ch06_seed.py` (below) checks for the column and will stop with the exact fix if it's missing; if you'd rather do it now, here it is verbatim from Chapter 4:

```sql
ALTER TABLE city_documents ADD COLUMN search_vector tsvector;

UPDATE city_documents
SET    search_vector = to_tsvector('english', title || ' ' ||
body);

CREATE OR REPLACE FUNCTION city_documents_search_vector_update()
RETURNS trigger AS $$
BEGIN
    NEW.search_vector := to_tsvector('english', NEW.title || ' '
|| NEW.body);
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_city_documents_search_vector
    BEFORE INSERT OR UPDATE OF title, body ON city_documents
    FOR EACH ROW
    EXECUTE FUNCTION city_documents_search_vector_update();

CREATE INDEX idx_city_documents_search_vector
    ON city_documents USING GIN (search_vector);
```

Run the Chapter 6 seed

This adds the (initially empty) `embedding` column to `city_documents` and creates and populates `city_photos`. It needs only `psycopg`, `numpy`, and `pgvector` — not `sentence-transformers` — since `city_photos`’s vectors are synthetic:

```
python data/ch06_seed.py
```

Expected output:

```
Connecting to: dbname=portsmith
Applying DDL …
Generating 5000 synthetic photo embeddings …
Done — 5000 rows in city_photos across 10 categories.
city_documents.embedding added but left NULL — run `python data/
ch06_embed_documents.py` next.
```

Compute real embeddings for `city_documents`

This is the step that actually loads and runs the model:

```
python data/ch06_embed_documents.py
```

Expected output:

```
Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 (first run downloads ~90MB of model
weights) …
Embedding 30 documents …
Done — 30 rows in city_documents now have an embedding.
```

Verify the load

Open `psql portsmith` and run these checks.

Check 1 — `city_photos` structure:

```
\d city_photos
```

```
                                Table "public.city_photos"
    Column     |    Type     | Collation | Nullable
|                Default
---------------+-------------+-----------+----------
+-----------------------------------------
 id            | integer     |           | not null |
nextval('city_photos_id_seq'::regclass)
 category      | text        |           | not null |
 neighbourhood | text        |           | not null |
 caption       | text        |           | not null |
 embedding     | vector(384) |           | not null |
Indexes:
    "city_photos_pkey" PRIMARY KEY, btree (id)
```

Check 2 — 500 photos per category:

```
SELECT category, COUNT(*) FROM city_photos GROUP BY category ORDER
BY category;
```

```
          category           | count
-----------------------------+-------
 community_event             |   500
 harbour_waterfront          |   500
 historic_architecture       |   500
 industrial_dock             |   500
 infrastructure_construction |   500
 municipal_building          |   500
 public_park                 |   500
 residential_street          |   500
 street_market               |   500
 wildlife_nature             |   500
(10 rows)
```

Check 3 — every `city_documents` row has an embedding:

```
SELECT COUNT(*) AS total, COUNT(embedding) AS with_embedding FROM
city_documents;
```

```
 total | with_embedding
-------+----------------
    30 |             30
```

If all three match, proceed to the exercises.

Exercises

Exercise 1 — The `vector` Type

1.1 — Literals and dimensions

A `vector` literal is a bracketed list of numbers:

```sql
SELECT '[1,2,3]'::vector;
```

```
 vector
---------
 [1,2,3]
```

A `vector(384)` column, like `city_documents.embedding`, enforces its dimension count at write time — every row's vector must have exactly 384 numbers. There is no meaningful way to compare a 384-dimensional embedding to a 3-dimensional one, so PostgreSQL simply won't let a mismatched value in:

```sql
SELECT '[1,2,3]'::vector(384);
```

```
ERROR:  expected 384 dimensions, not 3
```

1.2 — What's actually in the column

`city_documents.embedding` isn't hand-written — it came from `ch06_embed_documents.py` calling `all-MiniLM-L6-v2` on each document's title and body. Look at a real one:

```sql
SELECT id, title, embedding FROM city_documents WHERE id = 1;
```

The `embedding` value prints as 384 comma-separated floating-point numbers between roughly -1 and 1 — not something a human reads directly, but exactly what every operator and index in this chapter operates on. What those particular 384 numbers *mean* isn't individually interpretable (no single dimension corresponds to a concept like “is about harbours”); what matters is only their *position relative to other documents’ vectors*, which is what the distance operators in Exercise 2 measure.

Exercise 2 — Exact Nearest Neighbors: `<->`, `<#>`, `<=>`

`pgvector` gives you three distance operators. They agree more often than you’d expect, and the case where they disagree is worth understanding before it surprises you in production.

2.1 — Three tiny vectors, three operators

```sql
SELECT '[1,0,0]'::vector <-> '[0,1,0]'::vector AS l2_distance;
```

```
     l2_distance
--------------------
 1.4142135623730951
```

`<->` is **Euclidean (L2) distance** — straight-line distance between the two points, exactly like distance in ordinary geometry. Two orthogonal unit vectors are √2 apart.

```sql
SELECT '[1,0,0]'::vector <#> '[0,1,0]'::vector AS
neg_inner_product;
```

```
 neg_inner_product
--------------------
                 -0
```

`<#>` is the **negative inner product** — the dot product of the two vectors, negated (pgvector negates it so that, consistent with the other two operators, *smaller means closer*). Orthogonal vectors have a dot product of zero.

```sql
SELECT '[1,0,0]'::vector <=> '[0,1,0]'::vector AS cosine_distance;
```

```
 cosine_distance
------------------
                1
```

`<=>` is **cosine distance** (`1 - cosine similarity`) — the angle between the two vectors, completely ignoring their length. Orthogonal vectors have maximum cosine distance, `1`.

2.2 — Where they disagree: magnitude

The three operators can rank the *same* pair of candidates in a *different* order the moment the vectors involved aren't the same length. Compare a query vector against two candidates — one that points in exactly the same direction but is twice as long, and one that points in a slightly different direction but is almost the same length:

```
SELECT
  '[1,0]'::vector <-> '[2,0]'::vector   AS l2_to_a,
  '[1,0]'::vector <-> '[1,0.5]'::vector AS l2_to_b,
  round(('[1,0]'::vector <=> '[2,0]'::vector)::numeric, 4)   AS
cos_to_a,
  round(('[1,0]'::vector <=> '[1,0.5]'::vector)::numeric, 4) AS
cos_to_b;
```

```
 l2_to_a | l2_to_b | cos_to_a | cos_to_b
---------+---------+----------+----------
       1 |     0.5 |   0.0000 |   0.1056
```

The ranking flips. By `<->` (L2), candidate B (`[1,0.5]`, distance `0.5`) is closer than candidate A (`[2,0]`, distance `1`). By `<=>` (cosine), it's the reverse — A is distance `0` (identical direction) while B is `0.1056` away. Neither operator is "wrong"; they're answering different questions. L2 asks "how far apart are these points." Cosine asks "how similarly are these vectors oriented, regardless of scale."

2.3 — Why this chapter normalizes, and what that buys you

`ch06_embed_documents.py` calls `model.encode(texts, normalize_embeddings=True)` — every stored embedding is scaled to unit length before it's written. Once every vector in a comparison has the same length, the magnitude difference that caused Exercise 2.2's disagreement simply can't occur, and all three operators produce the *same ranking* (just on different numeric scales). Embedding the query "waterfront redevelopment funding" (Exercise 5 shows how) and checking the top 3 `city_documents` rows by each operator confirms it:

	`<->` (L2)	**`<#>` (neg. inner product)**	**`<=>` (cosine)**
Rank 1	doc 1 — `0.8605`	doc 1 — `-0.6298`	doc 1 — `0.3702`
Rank 2	doc 22 — `1.0546`	doc 22 — `-0.4439`	doc 22 — `0.5561`
Rank 3	doc 5 — `1.0646`	doc 5 — `-0.4333`	doc 5 — `0.5667`

Same three documents, same order, every time — only the numbers' scale differs between columns. The practical rule: **normalize your embeddings, and pick whichever operator has an index you can build** — for normalized vectors it stops being a search-quality decision and becomes a performance one, covered next.

Exercise 3 — IVFFlat: Approximate Search with `lists` and `probes`

3.1 — Exact search doesn't scale

`city_photos` has 5,000 rows — small by real standards, but already enough to feel the cost of brute-force search. Find the 10 nearest neighbors of photo `id = 1` by comparing it against every other row:

```
        EXPLAIN (ANALYZE, BUFFERS)
        SELECT id, category, embedding <-> (SELECT embedding FROM
city_photos WHERE id = 1) AS dist
        FROM   city_photos
        WHERE  id != 1
        ORDER  BY embedding <-> (SELECT embedding FROM city_photos WHERE
id = 1)
        LIMIT  10;
```

```
                                                          QUERY PLAN
------------------------------------------------------------------
 Limit  (cost=1439.32..1439.35 rows=10 width=30) (actual
time=4.164..4.167 rows=10 loops=1)
   ->  Sort  (cost=1431.02..1443.52 rows=4999 width=30) (actual
time=4.163..4.164 rows=10 loops=1)
         Sort Key: ((city_photos.embedding <-> $0))
         Sort Method: top-N heapsort  Memory: 26kB
         ->  Seq Scan on city_photos  (cost=0.00..1323.00
rows=4999 width=30) (actual time=0.024..3.381 rows=4999 loops=1)
```

Every one of the 4,999 other rows gets its distance computed and sorted — about 4.6ms here. That cost grows linearly with table size; at real photo library scale (millions of rows) it stops being viable.

3.2 — Build the index

```
        CREATE INDEX idx_city_photos_ivfflat
            ON city_photos
            USING ivfflat (embedding vector_l2_ops)
            WITH (lists = 50);
```

```
CREATE INDEX
Time: 152.129 ms
```

`lists = 50` partitions the 5,000 vectors into 50 clusters (via k-means at index-build time) and stores each vector under its nearest cluster centroid. A rule of thumb from the pgvector docs: `lists ≈ rows / 1000` for up to roughly a million rows.

3.3 — The planner picks it up on its own

```
        EXPLAIN (ANALYZE, BUFFERS)
        SELECT id, category, embedding <-> (SELECT embedding FROM
city_photos WHERE id = 1) AS dist
        FROM   city_photos
        WHERE  id != 1
        ORDER  BY embedding <-> (SELECT embedding FROM city_photos WHERE
id = 1)
        LIMIT  10;
```

```
                                                              QUERY
PLAN
------------------------------------------------------------------
 Limit  (cost=61.55..62.35 rows=10 width=30) (actual
time=0.145..0.168 rows=10 loops=1)
   ->  Index Scan using idx_city_photos_ivfflat on city_photos
(cost=53.25..455.00 rows=4999 width=30) (actual time=0.145..0.166
rows=10 loops=1)
         Order By: (embedding <-> $0)
```

No `enable_seqscan = off` trick needed this time, unlike the small tables in earlier chapters — at 5,000 rows the planner's own cost estimate already favors the index: **4.6ms down to about 0.2ms.**

3.4 — The catch: default `probes` gives up a lot of accuracy

An IVFFlat search only examines the clusters nearest the query vector, not all 50 — controlled by `ivfflat.probes`, which **defaults to 1**. Compare the approximate top-10 against the true exact top-10:

```
        -- Exact (force a sequential scan to get true ground truth):
        SET enable_indexscan = off;
        SELECT array_agg(id) FROM (
            SELECT id FROM city_photos WHERE id != 1
            ORDER BY embedding <-> (SELECT embedding FROM city_photos
WHERE id = 1) LIMIT 10
        ) t;
```

```
                 array_agg
--------------------------------------------
 {146,357,485,468,500,261,384,397,185,226}
```

```
        -- Approximate, default probes = 1:
        RESET enable_indexscan;
        SELECT array_agg(id) FROM (
```

```
            SELECT id FROM city_photos WHERE id != 1
            ORDER BY embedding <-> (SELECT embedding FROM city_photos
WHERE id = 1) LIMIT 10
        ) t;

                 array_agg
--------------------------------------------
 {485,261,185,226,385,22,274,433,56,327}
```

Only **4 of the 10** results match (`485`, `261`, `185`, `226`) — **40% recall**, at the default setting, on a real query. This is not a contrived worst case; it's what `ivfflat.probes = 1` actually does: it searches only the single cluster closest to the query and simply never looks at the other 49, even though some of the true nearest neighbors live in a different cluster.

> **A note on reproducing these exact numbers:** IVFFlat's clustering step uses k-means with a random initialization at *index build* time — even against `city_photos`'s fixed, seeded synthetic data, rebuilding the same `CREATE INDEX ... USING ivfflat` statement can shift which photos land in which of the 50 clusters, and therefore which specific IDs and exact recall percentage `probes = 1` produces. Rerunning this exercise on your own machine will very likely show different IDs and a different percentage than the ones printed here. The pattern that reproduces reliably is the *shape*: `probes = 1` is measurably, often substantially, incomplete.

3.5 — Raise `probes`, watch recall recover

```
        SET ivfflat.probes = 2;

                 array_agg
--------------------------------------------
 {146,357,485,468,500,261,384,397,185,226}
```

At `probes = 2` the result already matches the exact set exactly — **100% recall** on this query. Timing at a slightly more conservative `probes = 5`, and at `probes = 50` (every list — the most exhaustive an IVFFlat search can be):

`probes`	Recall (this query)	Query time
1 (default)	40%	~0.4ms
5	100%	~0.6ms
50 (all lists)	100%	~0.6ms

Timings vary run to run more than the recall figures do (expect noise in the tenths of a millisecond on a table this size) — the shape that matters is that `probes = 1` is consistently fastest *and* the least reliable, while anything from `probes = 5` up delivers full recall on this query without a further, proportional cost from probing still more lists; the IVFFlat index structure keeps even an “examine every list” search meaningfully faster than the 4.6ms plain sequential scan from Exercise 3.1. Reset before continuing:

```
SET ivfflat.probes = 1;
```

Two probes happened to be enough for this particular query; it isn’t a guarantee for every query, which is why a small safety margin (a starting point of `probes ≈ √lists`, so `√50 ≈ 7`) is the honest recommendation rather than the bare minimum that worked once. The overall lesson: **IVFFlat’s out-of-the-box defaults are fast and unreliable at the same time** — `lists` and `probes` are not optional tuning, they’re the whole feature.

Exercise 4 — HNSW: A Different Trade-off

4.1 — Build it, and time the build

```
CREATE INDEX idx_city_photos_hnsw
    ON city_photos
    USING hnsw (embedding vector_l2_ops);
```

```
CREATE INDEX
Time: 513.963 ms
```

Compare against IVFFlat’s build time from Exercise 3.2: **152ms.** HNSW (Hierarchical Navigable Small World) builds a multi-layer graph structure connecting each vector to its approximate neighbors, which costs roughly **3x longer to build** than IVFFlat’s k-means clustering, on this table.

4.2 — Recall at HNSW’s default settings

Drop (or temporarily disable) the IVFFlat index so the planner has no choice but to use HNSW, then repeat the exact same top-10 query from Exercise 3.4:

```
BEGIN;
DROP INDEX idx_city_photos_ivfflat;

SELECT array_agg(id) FROM (
    SELECT id FROM city_photos WHERE id != 1
```

```
            ORDER BY embedding <-> (SELECT embedding FROM city_photos
WHERE id = 1) LIMIT 10
        ) t;

        ROLLBACK;  -- put idx_city_photos_ivfflat back

                 array_agg
--------------------------------------------
 {146,357,485,468,500,261,384,397,185,226}
```

Exact match with the true top-10 — 100% recall, at HNSW's default settings, no tuning required. Query time for this was about 0.63ms. Repeating the comparison against a second, independent query point (`id = 1001`, a `public_park` photo) confirms it isn't a fluke: HNSW again returned the exact top-10, while IVFFlat at default `probes = 1` again recovered only 3 of 10 (30% recall — as Exercise 3.4 noted, the exact percentage shifts between index rebuilds; what doesn't shift is that it's consistently well short of complete). HNSW's default, across every test in this chapter, has not missed a single true neighbor.

4.3 — The trade-off, stated plainly

	IVFFlat	HNSW
Build time (5,000 rows)	152ms	514ms (~3x)
Recall at defaults	30-40% in this chapter's tests, varies by build (needs `probes` tuned up)	~100% (no tuning needed)
Query time (well-tuned)	Comparable	Comparable
Tuning burden	On you, every session (`probes` is a session-level `SET`)	Minimal

This matches the general guidance in the pgvector documentation: **HNSW is the better default choice for most workloads** — better recall out of the box, no per-session parameter to remember to set. IVFFlat's advantages are real but narrower: faster to build (relevant if you rebuild the index often, e.g. after large bulk loads) and somewhat lower memory overhead at very large scale. Choose IVFFlat deliberately for those reasons, not by default.

Exercise 5 — Semantic Search: Finding Meaning, Not Words

5.1 — A query with no vocabulary overlap

`data/ch06_semantic_search.py` embeds a query string with the same model used to build the column, and searches `city_documents` by cosine similarity:

```
        python data/ch06_semantic_search.py "stray dogs and pet-friendly
spaces"

  id      sim  title
   6   0.4447  Council Minutes — Riverside Dog Park Funding
  24   0.3927  Public Notice — Riverside Dog Park Ribbon-Cutting
Event
  16   0.2683  Zoning Ordinance — Riverside Short-Term Rental
Restrictions
  ...
```

Not one word of the query — "stray," "dogs," "pet-friendly," "spaces" — appears in document 6's title, and full-text search (Chapter 4) would return zero rows for it; there's no shared lexeme to match on, misspelled or otherwise, so Chapter 5's trigram matching wouldn't help either. The embedding model has no notion of exact words at all — it read "the city's dog park" and "somewhere my pet can run off-leash" as *about the same thing*, which is precisely the gap described in this chapter's Background.

5.2 — A stronger example

```
        python data/ch06_semantic_search.py "waterfront redevelopment
funding"

  id      sim  title
   1   0.6298  Council Minutes — Harbour District Waterfront
Renovation Budget
  22   0.4439  Public Notice — Public Hearing on Harbour District
Waterfront Rezoning
   5   0.4333  Council Minutes — Riverside Road Resurfacing
Program
  12   0.4207  Zoning Ordinance — Harbour District Waterfront
Height Variance
   9   0.4101  Council Minutes — Canal Road Bike Lane Expansion
  ...
```

5.3 — Reading the gap, not just the ranking

Look at the jump between rank 1 (`0.6298`) and rank 2 (`0.4439`) — a much bigger drop than between ranks 2 through 5. That gap is meaningful: it says document 1 isn't merely the *best available* match, it's a *genuinely strong* one, while the rest are progressively looser associations. Semantic search has a property full-text search doesn't: **it never returns zero results.** Every document has *some* cosine similarity to any query — the model will confidently rank all thirty documents even for a nonsense query, it just won't find any of them close. A real application needs a similarity-score cutoff (or a "confidence" gap check like this one) to distinguish "found it" from "here's our least-bad guess," a problem keyword search's honest zero-rows sidesteps entirely.

Exercise 6 — Hybrid Search: Blending Keyword and Semantic Signals

Semantic search and keyword search fail differently. Semantic search misses exact terms that matter — a specific ordinance number, a street name — treating them as just more words to average into a general impression of meaning. Keyword search misses paraphrases entirely, as Exercise 5 just showed. Hybrid search runs both and blends the scores.

6.1 — Keyword-only ranking

```
        SELECT id, title, round(ts_rank(search_vector,
plainto_tsquery('english','bike lane'))::numeric,4) AS kw_rank
        FROM   city_documents
        WHERE  search_vector @@ plainto_tsquery('english','bike lane')
        ORDER  BY kw_rank DESC;

 id |
title                                | kw_rank
----
+------------------------------------------------------------------
+---------
  9 | Council Minutes — Canal Road Bike Lane
Expansion                    |  0.2928
 28 | Public Notice — Canal Road Bike Lane Construction
Schedule          |  0.1960
 30 | Public Notice — Annual Fireworks Display and Street
Closures        |  0.1759
  5 | Council Minutes — Riverside Road Resurfacing
Program                 |  0.0992
 15 | Zoning Ordinance — Reduced Parking Minimums Near Transit
Corridors |  0.0991
```

6.2 — Semantic-only ranking, same query

```
python data/ch06_semantic_search.py "bike lane" --top 8
```

```
  id      sim  title
   9   0.5513  Council Minutes — Canal Road Bike Lane Expansion
  28   0.4433  Public Notice — Canal Road Bike Lane Construction
Schedule
  15   0.4117  Zoning Ordinance — Reduced Parking Minimums Near
Transit Corridors
   5   0.3965  Council Minutes — Riverside Road Resurfacing
Program
  21   0.2754  Public Notice — Bay Street Water Main Repair Road
Closure
   6   0.2688  Council Minutes — Riverside Dog Park Funding
  30   0.2653  Public Notice — Annual Fireworks Display and Street
Closures
  10   0.2359  Council Minutes — Special Session on Flood
Mitigation Infrastructure
```

Both agree on the top 2 (documents 9 and 28 — a real, strong signal either way finds them). They disagree further down: document 30 ranks **3rd** by keyword (it happens to use the phrase "biking along the Canal Road bike lane" once, in passing) but only **7th** by semantic similarity, while document 15 ranks **5th** by keyword but **3rd** semantically (it's substantively about bike-lane-adjacent parking policy, using different phrasing throughout).

6.3 — Blend both into one score

```
python data/ch06_semantic_search.py "bike lane" --hybrid --top 8
```

```
  id       kw      sem    hybrid  title
   9   0.2928   0.5513   0.4479  Council Minutes — Canal Road Bike
Lane Expansion
  28   0.1960   0.4433   0.3444  Public Notice — Canal Road Bike
Lane Construction Schedule
  15   0.0991   0.4117   0.2867  Zoning Ordinance — Reduced
Parking Minimums Near Transit Corridors
   5   0.0992   0.3965   0.2776  Council Minutes — Riverside Road
Resurfacing Program
  30   0.1759   0.2653   0.2295  Public Notice — Annual Fireworks
Display and Street Closures
  21   0.0000   0.2754   0.1652  Public Notice — Bay Street Water
Main Repair Road Closure
   6   0.0000   0.2688   0.1613  Council Minutes — Riverside Dog
Park Funding
  10   0.0000   0.2359   0.1415  Council Minutes — Special Session
on Flood Mitigation Infrastructure
```

The underlying SQL (`HYBRID_SQL` in `ch06_semantic_search.py`) is a `LEFT JOIN` between a keyword-matched candidate set and a semantic top-10 candidate set, so documents 21, 6, and 10 — never matched by the keyword query at all (`kw = 0.0000`) — still appear, carried in purely by semantic similarity:

```
        hybrid_score = 0.4 * COALESCE(kw_score, 0) + 0.6 * COALESCE(sem_score,
0)
```

Document 15 moves from 5th (keyword) up to 3rd (hybrid); document 30 moves from 3rd (keyword) down to 5th (hybrid) — the blend genuinely changes the ranking, not just the score.

6.4 — The honest caveat about the weights

`0.4` and `0.6` here are fixed constants chosen because `ts_rank` and cosine similarity happen to land in roughly comparable numeric ranges on *this* dataset — they are not principled, and they are not portable to a different corpus or a different embedding model without re-checking that assumption. A production hybrid-search system typically normalizes each signal (min-max or z-score, over the actual candidate set returned) before blending, rather than trusting that two arbitrary scoring functions happen to land in the same numeric neighborhood. Treat the weights here as a teaching simplification, not a formula to copy into production untested.

Summary — What You Should Now Know

Tool	What it does
`vector(N)`	A fixed-dimension floating-point array column type; enforces `N` at write time
`<->`	L2 (Euclidean) distance — straight-line distance between points
`<#>`	Negative inner product — dot product, negated so smaller means closer
`<=>`	Cosine distance — angle between vectors, ignoring magnitude
`normalize_embeddings=True`	Makes all three operators agree on ranking; without it, they can disagree
`USING ivfflat (col vector_l2_ops) WITH (lists = N)`	Clusters vectors into `N` buckets; fast to build, needs `probes` tuned up from its default to get usable recall

Tool	What it does
`SET ivfflat.probes = N`	How many clusters an IVFFlat search actually examines — defaults to `1`, often too low
`USING hnsw (col vector_l2_ops)`	Graph-based ANN index; ~3x slower to build than IVFFlat here, but near-exact recall by default
`ts_rank(...) + (1 - cosine_distance)` weighted blend	Hybrid search — keyword precision plus semantic recall in one ranking

The key design insight from this chapter is that “approximate” in “approximate nearest neighbor” is not a rounding error — at IVFFlat’s default settings, this chapter repeatedly measured 30-40% recall on real queries, which means 6-7 of the top 10 *true* nearest neighbors were silently missing from the results, with no error, no warning, nothing to indicate anything was wrong. Every ANN index trades recall for speed on a dial you control (`probes`, `ef_search`); the failure mode to avoid is not knowing where that dial is set.

The `city_documents` table’s `embedding` column is reused in Chapter 10 (PostgREST), which exposes the semantic search query from Exercise 5 as a public RPC endpoint alongside the fuzzy “did you mean?” search from Chapter 5.

Going further: this chapter used `all-MiniLM-L6-v2` (384 dimensions) because it’s small, fast on CPU, and good enough to teach with — production systems often use larger models (OpenAI’s `text-embedding-3-small` at 1536 dimensions, or similar) for better semantic quality, at proportionally higher storage and compute cost per vector. For tables much larger than `city_photos`’s 5,000 rows, revisit `lists` (scale roughly with row count) and consider `hnsw`’s `m` and `ef_construction` build-time parameters, which trade index size and build time for recall the same way `probes` trades query time for it. And if your embeddings change meaning over time — a newer, better model version — remember that a `vector` column has no built-in versioning; re-embedding the whole table and rebuilding the index is a full migration, not an in-place update.

Bonus Section — Build a Local RAG System

*This section is optional and sits outside the numbered exercises — nothing later in the book depends on it. It exists because semantic search (Exercise 5) is one step short of the thing most people actually mean when they say “AI search”: a system that doesn’t just return matching documents, but reads them and answers the question directly. That’s **Retrieval- Augmented Generation (RAG)**, and you already built the hard half of it — retrieval — in Exercises 1 through 5. This section adds the other half.*

What RAG actually is

Every large language model has a knowledge cutoff and no idea what’s in your `city_documents` table. Ask one directly “what are the rules for Portsmith’s dog park?” and it will either say it doesn’t know or, worse, confidently invent something plausible-sounding and wrong — a **hallucination**. RAG sidesteps this without retraining or fine-tuning anything: before asking the model the question, retrieve the most relevant real text from your own database and paste it directly into the prompt, instructing the model to answer *only* from that text. The model isn’t recalling facts from training anymore; it’s reading comprehension over text you handed it seconds ago.

This means RAG is two independent systems wired together, and it’s worth keeping them mentally separate:

1. **Retrieval** — PostgreSQL, `pgvector`, and everything from Exercises 1-5 of this chapter. Given a question, find the most relevant chunks of text.
2. **Generation** — a large language model that turns “here are some relevant facts, here is a question” into a fluent, direct answer.

Nothing about retrieval changes when you add generation on top of it — the <=> cosine-distance query from Exercise 5 is exactly what runs here too. What’s new is chunking (documents are usually too long and too topically mixed to embed as a single vector) and the LLM call itself, which this section runs locally through **Ollama** rather than a paid API, for the same reason Chapter 6 as a whole runs its embedding model locally: no API key, no per-query cost, nothing leaves your machine.

Installation — Ollama

```
curl -fsSL https://ollama.com/install.sh | sh
```

This installs Ollama as a systemd service listening on `localhost:11434`. Pull a chat model — this section uses `llama3.1:8b` (about 4.9GB):

```
ollama pull llama3.1:8b
```

If disk space or RAM is tight, `ollama pull phi` gets you a much smaller (~1.6GB) model at a real quality cost; every example below works with either, since the model name is a plain argument to the RAG script, not something hardcoded.

> **Note:** the *first* request to any given model incurs a one-time load delay (10-15 seconds isn't unusual) while Ollama loads its weights into memory. Every request after that, to the same model, is fast — Ollama keeps recently used models resident for a while. Don't judge a model's real latency by its very first response.

Ingest: chunk and embed a directory of documents

`data/rag_docs/` contains four new, longer, informational documents — a business licensing guide, a parks and recreation guide, a utilities guide, and a newcomer's guide — distinct from `city_documents`'s formal council records, written in the plain-prose register an actual city help-desk handout would use. `ch06_rag_ingest.py` splits each file into overlapping word-count chunks, embeds every chunk with `all-MiniLM-L6-v2` (the same model used everywhere else in this chapter — retrieval only works if the same model embeds both the documents and the questions), and loads them into a table you name on the command line:

```
        python data/ch06_rag_ingest.py data/rag_docs --table
portsmith_rag --recreate

Connecting to: dbname=portsmith
Loading all-MiniLM-L6-v2 …
  getting_a_business_license.txt: 4 chunks
  newcomer_guide.txt: 4 chunks
  parks_and_recreation.txt: 4 chunks
  utilities_and_public_works.txt: 4 chunks
Done — 16 chunks from 4 files loaded into 'portsmith_rag'.
```

Each ~700-word document became 4 chunks of 180 words with a 40-word overlap between consecutive chunks (both configurable via `--chunk-size` and `--chunk-overlap`). The overlap matters: without it, a sentence that happens to fall right at a chunk boundary gets split across two chunks, and neither half alone may retrieve well for a question about it.

The table it creates is deliberately the same generic shape regardless of what you point it at:

```
        \d portsmith_rag

                                 Table "public.portsmith_rag"
   Column    |    Type     | Collation | Nullable
|                  Default
------------+------------+-----------+----------
+-----------------------------------------
 id          | integer     |           | not null |
nextval('portsmith_rag_id_seq'::regclass)
 source      | text        |           | not null |
 chunk_index | integer     |           | not null |
 content     | text        |           | not null |
 embedding   | vector(384) |           | not null |
Indexes:
    "portsmith_rag_pkey" PRIMARY KEY, btree (id)
    "idx_portsmith_rag_embedding" hnsw (embedding
vector_cosine_ops)
```

`--table` is a plain identifier you choose — point the same script at a different directory with `--table` set to something else, and you have a second, independent knowledge base in the same database. Run it against your own notes, a project's documentation, anything in `.txt` or `.md` files.

> **Why the table name is validated, not just interpolated:** `--table` comes from the command line and flows into `CREATE TABLE`, `INSERT`, and `CREATE INDEX` statements. SQL identifiers (table names) can't be passed as query parameters the way values can — `cur.execute("... WHERE id = %s", (table_name,))` only works for *values*. `ch06_rag_ingest.py` and `ch06_rag_chat.py` both reject anything that isn't a plain lowercase identifier before it touches SQL, and use `psycopg.sql.Identifier` to quote it correctly rather than dropping it into an f-string. For a script you run yourself against your own database this is a small concern; the habit is worth keeping anyway, because the same shortcut in a script that takes a table name from a web form is a real SQL injection hole.

Ask a question

`ch06_rag_chat.py` takes three required arguments — `model`, `table`, `question`, in that order — plus `--host` and `--port` for Ollama (both default to `localhost` and `11434`, so you only need them if Ollama runs elsewhere):

```
python data/ch06_rag_chat.py llama3.1:8b portsmith_rag \
    "How do I get a business license in Portsmith?" --show-context
```

```
--- retrieved context ---
[getting_a_business_license.txt#0  dist=0.2743] Getting a Business
License in Portsmith Anyone opening a business within city limits
needs a busines...
[getting_a_business_license.txt#3  dist=0.5796] to change their
registered category — a retail shop adding a small cafe counter,
for example — need ...
[getting_a_business_license.txt#2  dist=0.6149] scratch rather
than simply renewing, which means a new round of inspections for
categories that requ...
[newcomer_guide.txt#0  dist=0.6176] A Newcomer's Guide to
Portsmith Welcome to Portsmith. This guide covers the basics every
new residen...

To get a business license in Portsmith, you need to start online
through the city's permitting portal, where you choose a business
category since it determines which inspections and follow-up
permits apply. Most applications are processed within two to three
weeks, but those involving food service or alcohol take longer
because they require scheduled in-person inspections.
```

`--show-context` prints exactly what got retrieved before the model ever saw the question — the same top-K cosine-distance query from Exercise 5, just with `content` instead of `title` as the payload. The answer isn't copied from any single chunk; it's the model synthesizing across the top few, which is the actual value RAG adds over raw semantic search: Exercise 5 would have handed a person four ranked chunks to read themselves, this hands them one sentence that already did the reading.

> **Reproducibility, honestly:** the retrieved chunks and their `dist` values above are fully deterministic — rerun the exact query and you'll get the exact same four chunks in the exact same order, because embedding and cosine distance are pure math. The generated *answer* is not — Ollama samples from the model's output distribution, so wording will differ run to run (sometimes prose, sometimes a bulleted list) even against identical context. That's a property of the generation half of RAG, not the retrieval half; don't expect to reproduce this exact paragraph verbatim.

What happens when the answer isn't in the documents

This is the case that actually matters for trusting a RAG system — ask something the ingested documents have no way of answering:

```
python data/ch06_rag_chat.py llama3.1:8b portsmith_rag \
    "What is the capital of France?" --show-context
```

```
--- retrieved context ---
[newcomer_guide.txt#1  dist=0.9228] river on the city's western
edge and is primarily residential, with the new dog park, several
parks ...
[utilities_and_public_works.txt#1  dist=0.9272] requires
temporarily shutting off service to a section of a neighbourhood,
Public Works issues a boi...
[newcomer_guide.txt#0  dist=0.9361] A Newcomer's Guide to
Portsmith Welcome to Portsmith. This guide covers the basics every
new residen...
[newcomer_guide.txt#2  dist=0.9475] alike. Noise ordinance
enforcement is more actively watched here than in other
neighbourhoods given ...

I don't know. The context only provides information about the city
of Portsmith, its neighbourhoods, and various services offered by
the city, but does not mention the capital of France.
```

Two things to notice. First, retrieval **still returned four chunks** — exactly as Exercise 5 described, cosine similarity search never returns "no results," it returns the *least bad* matches available, and here they're genuinely irrelevant. Second, look at the distances: `0.92-0.95`, versus `0.27-0.61` for the business license question. That gap is the signal a production system would actually act on — this script's prompt template happens to lean on the model's own judgment to say "I don't know," but a more defensive design would check the top distance against a threshold *before* even calling the LLM, and skip generation entirely for a question this poorly matched. Relying on the model to notice the context is irrelevant works here because `llama3.1:8b` is reasonably well-behaved about it — it is not a guarantee every model or every prompt will get right, and it's the single biggest reliability risk in any RAG system: nothing stops a language model from answering confidently from weak or irrelevant context if the prompt doesn't insist otherwise.

The prompt: the actual glue between the two systems

`ask_ollama()` in `ch06_rag_chat.py` sends the model one plain-text prompt via Ollama's `/api/generate` endpoint — no special "RAG mode" exists in Ollama or in the model itself, it's just a string:

```
You are a help-desk assistant for the city of Portsmith. Answer
the question using ONLY the context below. If the context doesn't
contain the answer, say you don't know rather than guessing.
```

```
Context:
(getting_a_business_license.txt) Getting a Business License in
Portsmith Anyone opening a business ...
(getting_a_business_license.txt) to change their registered
category — a retail shop adding a small ...
...

Question: How do I get a business license in Portsmith?

Answer:
```

This is the entire mechanism. There is no fine-tuning, no special API, nothing model-specific — "retrieval-augmented generation" is a prompt template plus a database query, which is exactly why it was worth showing you the whole thing at this level rather than reaching for a framework.

Going further: this section's retrieval is intentionally the simplest version that works — top-K by raw cosine distance, no re-ranking, no distance-threshold cutoff before generation, no limit on how many tokens of context get sent to the model regardless of length. Production RAG systems typically add a ***re-ranking*** *step (retrieve a wider candidate set cheaply, then re-score the top candidates with a slower, more accurate model before picking the final context),* ***hybrid retrieval*** *(blend `ts_rank` back in, exactly as Exercise 6 did, since keyword precision and semantic recall are just as complementary here as they were there), and explicit* ***citations*** *in the generated answer so a user can verify a claim against the source chunk rather than trusting the model's summary outright. None of that changes the shape of what you built here — it's still retrieval, then generation, with more care taken at each step.*